

Treatment Protocol Verification Report

AgroScan Bangladesh Agricultural Data Pack

Document type	Technical verification report
Scope	Crop disease advice, spray doses, PHI, and prevention protocols
Source dataset	disease_treatments.json (AgroScan BD pack)
Prepared for	Agricultural officer / DAE plant protection review
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Language	English (Bangla crop/disease names included)

I. Abstract

This report compiles disease-management protocols used by AgroScan for officer verification. Each entry lists identification keys, clinical summary, prevention, and chemical regimens with dose, interval, and pre-harvest interval (PHI). Reviewers should validate every product against the Plant Protection Wing (PPW) register and the printed label. Authorization appears only on the final page.

II. Verification method

- 1) Confirm disease identity and symptoms.
- 2) Confirm each chemical: active ingredient, dose, water volume, timing, interval, PHI.
- 3) Confirm registration on <https://ppw.krishi.gov.bd/pesticide-list>.
- 4) Record corrections beside the regimen table; sign and seal the final page only.

III. Disease protocols

Each subsection identifies one crop–disease protocol. Chemical regimens are tabulated for dose, spray interval (days), maximum seasonal applications, and PHI (days).

III.1 Apple — Apple scab

Bangla: আপলে — আপলের স্কাব রোগ. Key: Apple__Apple_scab. Pathogen: *Venturia inaequalis*. Severity: medium. Pack confidence: high.

A. Summary

Apple scab is a disease of apple orchards in temperate climates. Bangladesh does not grow apple commercially, so this class is included for model completeness only. If AgroScan returns this result for a Bangladeshi field photo, treat it as a probable misclassification and ask the farmer to retake the photo or identify the crop.

B. Symptoms

- 1) Olive-green to brown velvety spots with feathery indistinct edges on leaves
- 2) Spots later turn dark and the leaf may distort and drop
- 3) Rough, corky, cracked scab lesions on the fruit
- 4) Requires prolonged spring leaf wetness at 10-24°C — conditions Bangladesh does not provide for apple

C. Immediate actions

- 1) Check whether this is really apple: Apple is not grown commercially in Bangladesh. Look at the leaf shape and the plant. If it is not an apple tree, retake the photo of a clearer, well-lit leaf and run the scan again...
- 2) If you are unsure of the crop, call the helpline: Call 16123 or visit your Upazila Agriculture Office with a sample leaf. Correct identification of the crop comes before any treatment.
- 3) Do not spray on the basis of this result: Spraying for a disease of a crop you are not growing wastes money and exposes you to chemicals for no reason.

D. Prevention

- 1) Not applicable in Bangladesh — apple is not a commercial crop here
- 2) If you are growing apple experimentally in a hill area, contact BARI Horticulture Research Centre for guidance

E. Fertilizer note

Not applicable. If the crop was misidentified, identify it correctly first and follow the BARC FRG-2018 recommendation for the actual crop.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 if you are not sure what crop this is or what is wrong with it.

III.2 Apple — Black rot

Bangla: আপলে — আপলের ব্ল্যাক রট রোগ. Key: Apple__Black_rot. Pathogen: *Botryosphaeria obtusa*. Severity: medium. Pack confidence: high.

A. Summary

A disease of temperate apple orchards, not relevant to Bangladeshi farming. Included for model completeness. Treat a Bangladeshi prediction of this class as a probable misclassification.

B. Symptoms

- 1) Purple-bordered 'frog-eye' leaf spots with tan centres
- 2) Sunken black cankers on branches
- 3) Firm black rot on fruit with concentric rings

C. Immediate actions

- 1) Check whether this is really apple: Apple is not grown commercially in Bangladesh. Retake the photo of a clearer leaf and rescan.
- 2) Call the helpline if unsure: Call 16123 or take a sample leaf to your Upazila Agriculture Office.
- 3) Do not spray on this result: Do not treat a crop you are not growing.

D. Prevention

- 1) Not applicable in Bangladesh

E. Fertilizer note

Not applicable. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 if you are not sure what crop this is.

III.3 Apple — Cedar apple rust

Bangla: আপলে — আপলের সড়ির রাস্ট রোগ. Key: Apple__Cedar_apple_rust. Pathogen: Gymnosporangium juniperi-virginianae. Severity: low. Pack confidence: high.

A. Summary

This disease cannot occur in Bangladesh because its alternate host, Eastern red cedar, does not grow here. Included for model completeness only. Any prediction of this class from a Bangladeshi photo should be shown as a misclassification, not a diagnosis.

B. Symptoms

- 1) Bright orange-yellow circular spots on apple leaves with small black dots in the centre
- 2) Tube-like structures on the underside
- 3) Orange gelatinous galls on cedar trees in spring

C. Immediate actions

- 1) This result is not possible in Bangladesh: The disease needs a cedar host that does not grow here. Retake the photo of a clearer leaf in good light and rescan.
- 2) If the leaf has orange spots, it is something else: Orange or rust-coloured spots on a Bangladeshi leaf are much more likely to be maize rust, wheat rust, bean rust or a leaf-spot fungus. Tell the app the crop name if you can.
- 3) Call 16123 if unsure: Take a sample leaf to your Upazila Agriculture Office.

D. Prevention

- 1) Not applicable in Bangladesh

E. Fertilizer note

Not applicable. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 to identify what your crop and its problem actually are.

III.4 Banana — Sigatoka leaf spot

Bangla: কলা — কলার সগিটোকা পাতার দাগ রোগ. Key: Banana__Sigatoka_leaf_spot. Pathogen: Mycosphaerella musicola (yellow Sigatoka) / M. fijiensis (black Sigatoka). Severity: high. Pack confidence: medium.

A. Summary

Streaks and spots that start as thin yellow lines on the leaf and grow into large dark blotches with grey centres, killing the leaves from the bottom up. Fewer functional leaves means smaller bunches and premature ripening.

B. Symptoms

- 1) Thin yellow-green streaks running parallel to the leaf veins on the underside, first visible on leaf 3-4 from the top
- 2) Streaks widen into brown then dark brown-black spots with a grey centre and yellow halo
- 3) Spots merge; the leaf dries and hangs down
- 4) Only a few functional leaves remain at bunch filling, so bunches are small
- 5) Fruit ripens prematurely and unevenly, sometimes before harvest
- 6) Worst in the rainy season with high humidity and in dense, poorly drained plantations

C. Immediate actions

- 1) Cut off and remove the dead and heavily spotted leaves: Cut leaves with more than 50% spotting, drag them out of the plantation and burn or bury. Do not leave them lying between the plants — they keep releasing spores. This is the back...
- 2) Desucker to open the plantation: Keep only 2-3 healthy suckers per mat. Overcrowded mats hold humidity and the disease never stops.
- 3) Improve drainage: Open drains between rows. Waterlogged banana plantations get Sigatoka far worse.
- 4) Spray with a sticker or oil: Banana leaves are waxy and plain water-based sprays run off. Mix a wetting agent, or use a mineral oil emulsion, so the fungicide sticks.
- 5) Rotate fungicide groups: Sigatoka develops resistance quickly. Alternate mancozeb (contact) with a triazole or strobilurin — never use the same systemic twice in a row.

D. Prevention

- 1) Weekly deleafing of spotted leaves — the single most effective practice
- 2) Desuckering to 2-3 followers per mat
- 3) Adequate spacing (2 x 2 m or wider) so air moves
- 4) Open drainage channels between rows
- 5) Use clean, tissue-culture or disease-free suckers when establishing a new plantation
- 6) Do not plant a new banana block next to an old heavily infected one

E. Fertilizer note

Banana is a very heavy potassium feeder. Follow FRG-2018 and apply potash in splits through the growing period. Potassium-deficient plants lose leaves to Sigatoka far faster. Adequate nitrogen keeps new leaves emerging to replace those lost.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP + wetting agent Mancozeb	2.5 g per litre of water plus 1 ml/L wetting agent	14	6	15
2	(unverified) Propiconazole 25% EC Propiconazole	1 ml per litre of water plus a wetting agent	21	3	21

Regimen 1 — Timing: Preventive, every 14-21 days through the rainy season. | Water: Roughly 8-12 litres of spray solution per decimal in an established plantation | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not harvest for 15 days after the last spray.

Regimen 2 — Timing: Curative, when streaking is already present. Alternate with mancozeb; do not use two triazole sprays in a row. | Water: 8-12 litres of spray solution per decimal | Brands: Tilt 250 EC, Propicon 250 EC | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 if plants have fewer than 6 functional leaves at bunch emergence, if bunches are ripening prematurely, or if the disease continues despite deleafing and two sprays.

III.5 Bell pepper / Capsicum — Bacterial spot

Bangla: ক্যাপসিকাম / মর্চি মরিচ — ক্যাপসিকামেরে ব্যাকটেরিয়াজনিত দাগ রোগ। Key: Pepper_bell__Bacterial_spot. Pathogen: Xanthomonas spp.. Severity: medium. Pack confidence: medium.

A. Summary

A bacterial disease — fungicides do not cure it. Small dark greasy spots on leaves and raised scabby spots on the fruit. It spreads in warm wet weather and through infected seed, and copper plus dry foliage is the only real control.

B. Symptoms

- 1) Small dark brown to black spots with a yellow halo on the leaves, greasy or water-soaked when fresh
- 2) Spots may drop out leaving shot holes; heavy leaf drop follows
- 3) Raised, rough, scabby spots on the fruit that destroy market value
- 4) Elongated dark lesions on stems and leaf stalks
- 5) Worst at 24-30°C in wet weather or after overhead irrigation

C. Immediate actions

- 1) Do not expect a fungicide to cure it: Bacteria do not respond to fungicides. Only copper-based products slow the spread.
- 2) Stop overhead irrigation immediately: Switch to drip. In a polytunnel, also increase ventilation so leaves dry quickly.
- 3) Do not work in wet plants: Hands, clothes and tools spread bacterial spot faster than rain.

- 4) Remove badly infected plants and spotted fruit: Bag them and remove from the house or field.
- 5) Apply copper: Copper oxychloride 50% WP at 3 g/L every 7-10 days as a protectant on healthy tissue.
- 6) Never save seed from this crop: The bacteria are seed-borne.

D. Prevention

- 1) Certified seed; hot water treatment at 50°C for 25 minutes
- 2) Drip irrigation only
- 3) Rotate away from pepper, chilli and tomato for 2 years
- 4) Stake plants and mulch
- 5) Disinfect stakes, ties and tools between crops
- 6) Ventilate polytunnels every morning

E. Fertilizer note

Avoid excess nitrogen. Keep calcium and potassium adequate. Do not apply foliar fertiliser sprays while the disease is active — the spray water spreads the bacteria.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Copper oxychloride 50% WP Copper oxychloride	3 g per litre of water	7	5	7

[AP AP-354]

Regimen 1 — Timing: Every 7-10 days while warm wet weather continues. | Water: 3-4 litres of spray solution per decimal | Brands: Cupravit 50 WP, Champion 77 WP, Blue Copper | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if fruit is scabbing or more than 20% of plants are affected.

III.6 Brinjal (eggplant) — Bacterial wilt

Bangla: বগুন — বগুনের ঢলে পড়া রোগ। Key: Brinjal__Bacterial_wilt. Pathogen: Ralstonia solanacearum. Severity: high. Pack confidence: high.

A. Summary

The plant wilts suddenly in the middle of the day while the leaves are still green, then dies within days. It is a soil bacterium — there is no spray that cures it. The answer is grafted seedlings, drainage and rotation.

B. Symptoms

- 1) Sudden wilting of the whole plant during the hottest part of the day, recovering slightly at night at first, then permanent
- 2) Leaves stay GREEN while the plant wilts — this is the key difference from fungal wilts, where leaves yellow first
- 3) Cut the stem near the base: the vascular ring is brown and a sticky greyish ooze comes out
- 4) The ooze test: hang a cut stem piece in a glass of clear water; within minutes a milky-white thread streams down from the cut. That confirms bacterial wilt.
- 5) Roots and lower stem may be brown and rotten
- 6) Spreads in waterlogged, warm soil and moves with irrigation water

C. Immediate actions

- 1) Do the ooze test before doing anything else: Cut a wilted stem near the base and hang it in clear water. A milky white thread streaming down means bacterial wilt. If you skip this you will probably spend money on a fungicide...
- 2) Remove wilted plants with the soil around the roots: Dig out each wilted plant along with a spadeful of surrounding soil, put it in a bag and burn or bury it far from the field. Do not shake soil off in the field.
- 3) Stop water flowing from infected to healthy areas: The bacteria travel in irrigation and rain water. Block the flow from the infected patch. Irrigate the healthy part of the field first, the infected part last.
- 4) Do not spray a fungicide: No fungicide cures bacterial wilt. Copper gives only marginal help. Money spent on spraying here is money wasted.
- 5) Improve drainage now: Open drains and raise the beds. Waterlogging is the single biggest driver of bacterial wilt in Bangladesh.
- 6) Apply bleaching powder to the removed plant's spot: A handful of bleaching powder worked into the hole where the plant was removed reduces the bacterial load in that spot.

D. Prevention

- 1) Use grafted seedlings — brinjal scion grafted onto wilt-resistant rootstock (*Solanum torvum*, locally 'tit begun', or resistant tomato rootstock). This is the most effective solution available in Bangladesh and is promoted by DAE and BARI.
- 2) Rotate with rice for 2-3 years; flooded rice culture strongly reduces *Ralstonia* in the soil
- 3) Never rotate brinjal with tomato, potato or chilli — all are hosts
- 4) Plant on raised beds with open drains
- 5) Do not use flood irrigation on brinjal; use furrow irrigation and control the flow direction
- 6) Disinfect tools between plants when pruning or staking

E. Fertilizer note

Avoid excess nitrogen, which makes plants more susceptible. Use well-decomposed organic manure — fresh cowdung can increase wilt. Apply lime if the soil is acidic; *Ralstonia* is more damaging in acid soils. Follow FRG-2018 for brinjal.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Copper oxychloride 50% WP (soil drench) Copper oxychloride	3 g per litre of water, drenched around the base of healthy plants ne...	15	2	7

[AP AP-354]

Regimen 1 — Timing: Immediately after removing wilted plants, applied to the surrounding healthy plants. | Water: Approximately 15-20 litres of drench solution per decimal | Brands: Cupravit 50 WP, Champion 77 WP, Blue Copper | PHI note: Do not pick brinjal for 7 days after application.

G. Escalation

Call 16123 as soon as you confirm bacterial wilt with the ooze test. Ask your Sub-Assistant Agriculture Officer about grafted seedling sources — this is the practical long-term fix and DAE promotes it.

III.7 Brinjal (eggplant) — Brinjal shoot and fruit borer (BSFB)

Bangla: বগুন — বগুনের ডগা ও ফল ছদ্মিকারী পোকা. Key: Brinjal__Shoot_and_fruit_borer. Pathogen: *Leucinodes orbonalis* (an insect pest, not a disease). Severity: high. Pack confidence: high.

A. Summary

This is Bangladesh's most damaging brinjal pest, not a disease. The caterpillar bores into the shoot tip (which droops and dies) and into the fruit (which becomes unsellable). It is also the reason brinjal is one of the most heavily sprayed crops in the country — pheromone traps and weekly clipping cut that spraying dramatically.

B. Symptoms

- 1) Young shoot tips wilt, droop and dry up — the classic 'dead heart' look
- 2) A small round hole plugged with brown excreta on the shoot or fruit
- 3) Fruit looks fine outside but is tunnelled and full of frass inside when you cut it
- 4) Damaged fruit is rejected at the market
- 5) Small brown-white moths with a triangular resting posture, active at night
- 6) Present all season wherever brinjal is grown continuously

C. Immediate actions

- 1) Clip and destroy every wilted shoot: Cut each drooping shoot about 5 cm BELOW the wilted part (the caterpillar is inside, below the visible damage), collect them in a bag and burn or bury. Do this twice a week. This ...
- 2) Remove all bored fruit: Pick every fruit with a hole in it, including those on the ground. Each one contains a caterpillar that will become a moth.
- 3) Install pheromone traps: Put up 40-60 pheromone traps per acre (about 15-25 per bigha), at canopy height, and change the lure every 4-6 weeks. This is a proven Bangladesh recommendation and greatly reduce...
- 4) Only spray if damage stays above threshold: If more than 5% of shoots or fruit are damaged despite clipping and trapping, spray. Rotate insecticide groups every time.
- 5) Never spray just before picking: This pest tempts farmers into spraying a few days before harvest. That is exactly how residue problems happen. Respect the PHI without exception.

D. Prevention

- 1) Clip and destroy infested shoots twice weekly — the backbone of BSFB management
- 2) Pheromone traps at 40-60 per acre

- 3) Grow tolerant varieties: Jhumka, Shingnath, Nayantara and Uttara are noted as tolerant in Bangladesh
- 4) Intercrop with cowpea, maize or coriander to support natural enemies
- 5) Rotate with non-solanaceous crops
- 6) Do not grow brinjal continuously in the same area year-round — a break starves the pest

E. Fertilizer note

Avoid excess nitrogen — soft succulent shoots are exactly what the larvae prefer. Follow FRG-2018 for brinjal and keep potassium adequate.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Emamectin benzoate 5% SG Emamectin benzoate	0.5 g per litre of water (about 5 g per 10 litre sprayer load)	12	3	5
2	[unverified] Flubendiamide 20% WG Flubendiamide	0.25-0.5 g per litre of water; follow the label rate exactly	15	2	7

Regimen 1 — Timing: Only when damage exceeds 5% after clipping and trapping. Time it right after a picking so the PHI clears before the next picking. | Water: 3-4 litres of spray solution per decimal | Brands: Proclaim 5 SG, Emastar 5 SG | PHI note: Do not pick brinjal for 5 days after the last spray. Plan sprays immediately after a picking round.

Regimen 2 — Timing: When damage exceeds threshold; the Plantwise Bangladesh factsheet recommends application at label dose every 15 days in severe cases. | Water: 3-4 litres of spray solution per decimal | Brands: Belt 24 WG, Fame 480 SC | PHI note: Do not pick brinjal for 7 days after the last spray.

G. Escalation

Call 16123 if damage stays above 10% despite clipping, traps and two sprays — that usually means insecticide resistance and you need a different approach, not a stronger dose.

III.8 Brinjal (eggplant) — Phomopsis blight and fruit rot

Bangla: বগুন — বগুনের ফোমোপসিস ধ্বসা ও ফল পচা রোগ। Key: Brinjal__Phomopsis_blight. Pathogen: Phomopsis vexans. Severity: medium. Pack confidence: medium.

A. Summary

Grey-brown spots with pale centres on leaves and stems, and sunken soft rotten patches on the fruit that make it unsellable. It is seed-borne, so treated seed and clean nursery beds prevent most of it.

B. Symptoms

- 1) Small circular to oval grey-brown spots on leaves with pale grey centres and dark margins
- 2) Tiny black dots (pycnidia) in the centre of older spots
- 3) Dark sunken cankers on the stem near the base, sometimes girdling and killing the plant
- 4) On fruit: pale sunken soft patches that spread and turn the fruit into a soft rotten mass, often with black dots on the surface
- 5) Seedlings damp off or develop stem cankers in the nursery
- 6) Worse in warm humid weather and where seed was saved from an infected crop

C. Immediate actions

- 1) Pick off and remove all rotting fruit: Every rotting fruit is releasing spores onto the rest of the crop. Bag them and take them out of the field — do not drop them between the rows.
- 2) Cut out stem cankers: Where a canker is on a branch, cut the branch below it and remove it. If the canker girdles the main stem near the base, remove the whole plant.
- 3) Spray a protectant: Mancozeb at 2-2.5 g/L or carbendazim at 1 g/L, every 10-12 days, covering fruit and stems as well as leaves.
- 4) Stop overhead watering: Switch to furrow or drip. Wet fruit surfaces are what let the fruit rot phase start.
- 5) Do not save seed from this crop: The fungus is carried inside the seed. Saving seed guarantees the problem returns next year.

D. Prevention

- 1) Treat seed before sowing — carbendazim or Trichoderma at 4-5 g/kg seed, or hot water at 50°C for 30 minutes
- 2) Use certified seed from BADC or a licensed supplier
- 3) Raise seedlings in a clean, well-drained nursery bed; solarise the bed before sowing

- 4) Rotate away from brinjal for 2 years
- 5) Remove and destroy all crop residue after harvest
- 6) Stake plants so fruit does not touch the soil

E. Fertilizer note

Follow FRG-2018 for brinjal. Keep potassium adequate to firm up fruit skin, which resists rot better. Avoid excess nitrogen, which produces soft fruit and dense humid canopies.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	10	5	7
2	Mancozeb, Carbendazim 50% WP (unverified) Carbendazim	1 g per litre of water for foliar spray; 4-5 g per kg of seed for see...	12	2	10

[AP AP-176, AP-241]

Regimen 1 — Timing: Preventive, every 10-12 days from first fruit set in humid weather. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not pick brinjal for 7 days after the last spray. Brinjal is picked continuously, so plan sprays around the picking ...

Regimen 2 — Timing: Seed treatment before sowing; foliar spray at first symptoms, maximum twice, alternated with mancozeb. | Water: 3-4 litres of spray solution per decimal | Brands: Bavistin DF, Knowin 50 WP, Autostin 50 WDG | PHI note: Do not pick brinjal for 10 days after the last spray.

G. Escalation

Call 16123 if more than 10% of fruit is rotting, if plants are dying from stem cankers, or if seedlings are collapsing in the nursery.

III.9 Cherry — Powdery mildew

Bangla: চেরি — চেরির পাউডার মিল্ডিউ রোগ. Key: Cherry_(including_sour)___Powdery_mildew. Pathogen: Podosphaera clandestina. Severity: low. Pack confidence: high.

A. Summary

Cherry is not grown in Bangladesh. This class is included for model completeness only and should be treated as a misclassification signal in the Bangladesh deployment.

B. Symptoms

- 1) White powdery patches on leaves and shoots

C. Immediate actions

- 1) This result is not possible in Bangladesh: Cherry is not grown here. Retake the photo of a clearer leaf and rescan.
- 2) If the leaf has white powder, it is something else: White powder on a Bangladeshi leaf is much more likely to be powdery mildew of cucurbits, mango or beans. Tell the app the crop name if you can.
- 3) Call 16123 if unsure: Take a sample leaf to your Upazila Agriculture Office.

D. Prevention

- 1) Not applicable in Bangladesh

E. Fertilizer note

Not applicable. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 to identify your crop.

III.10 Chilli — Chilli anthracnose (ripe fruit rot / die-back)

Bangla: মরিচ — মরিচের অ্যানথ্রাকনোসিস (ফল পচা ও ডগা মরা) রোগ. Key: Chilli___Anthracnose. Pathogen: Colletotrichum capsici / C. gloeosporioides. Severity: high. Pack confidence: medium.

A. Summary

Sunken dark circular spots with rings of tiny black dots on ripening chillies, plus die-back of the shoot tips. It is the main reason chilli growers lose their crop at the ripe stage in Bangladesh, and it is seed-borne, so treated seed prevents most of it.

B. Symptoms

- 1) Small water-soaked spots on ripening fruit that become sunken, circular, dark and enlarge
- 2) Concentric rings of tiny black dots (acervuli) inside the lesion — the identifying feature
- 3) In wet weather, pink-orange spore masses ooze from the lesions
- 4) Die-back: shoot tips blacken and die downward from the top
- 5) Whole fruit shrivels and dries on the plant, becoming straw-coloured
- 6) Worst when fruit ripens during rainy or very humid weather

C. Immediate actions

- 1) Pick and remove every infected fruit: Each spotted fruit is producing millions of spores. Collect them in a bag — including fallen ones — and burn or bury deep. Doing this twice a week matters more than the spray.
- 2) Harvest ripe fruit promptly: Do not leave ripe chillies on the plant waiting for a better price. Ripe fruit left on the plant in humid weather is where anthracnose multiplies.
- 3) Cut back die-back shoots: Prune blackened shoot tips back into healthy green wood and remove the cuttings from the field.
- 4) Spray a protectant: Mancozeb at 2-2.5 g/L or copper oxychloride at 3 g/L every 10 days. Where the disease is established, use a difenoconazole or azoxystrobin product once, then return to protectants.
- 5) Do not save seed: The fungus lives inside the seed. Saving seed from this crop guarantees the same disease next season.

D. Prevention

- 1) Treat seed before sowing — carbendazim at 2-3 g/kg or Trichoderma at 4-5 g/kg, or hot water at 52°C for 30 minutes
- 2) Use certified seed from BADC, Lal Teer, ACI or another licensed supplier
- 3) Rotate away from chilli, tomato and brinjal for 2 years
- 4) Plant on raised beds with good drainage; avoid waterlogging
- 5) Space plants for air movement and mulch the beds
- 6) Harvest ripe fruit every 5-7 days without fail

E. Fertilizer note

Follow FRG-2018 for chilli. Keep potassium and calcium adequate — they firm up the fruit wall and reduce infection. Avoid excess nitrogen, which produces soft fruit and a dense humid canopy.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	10	5	7
2	Mancozeb Difenoconazole 25% EC Difenoconazole	0.5 ml per litre of water	12	2	7

[AP AP-384]

Regimen 1 — Timing: Preventive, every 10 days from first fruit set through the ripening period, especially in wet weather. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not pick for 7 days after the last spray. Green chilli is eaten fresh, so this matters.

Regimen 2 — Timing: Curative, when lesions are already on the fruit. Maximum 2 per season, alternated with mancozeb. | Water: 3-4 litres of spray solution per decimal | Brands: Score 250 EC, Difen 250 EC | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if more than 15% of ripening fruit is affected, if shoot die-back is spreading, or if the same field gets it every year.

III.11 Chilli — Chilli leaf curl (virus complex + mite/thrips damage)

Bangla: মরচি — মরচিরে পাতা কঁকড়াণো রোগ. Key: Chilli__Leaf_curl. Pathogen: Chilli leaf curl begomovirus complex (whitefly-transmitted), often combined with yellow mite (Polyphagotarsonemus latus) and thrips damage. Severity: high. Pack confidence: medium.

A. Summary

Leaves curl, crinkle and bunch up and the plant stops growing. In Bangladesh this is usually a combination of a whitefly-borne virus and damage from yellow mite and thrips. Which one it is decides the treatment, so look closely before you spray.

B. Symptoms

- 1) VIRUS: leaves curl UPWARD, become small, thick and leathery, plant is stunted, internodes short, little or no fruit; whiteflies present

- 2) YELLOW MITE: leaves curl DOWNWARD along the edges, become narrow, elongated and shiny/bronzed underneath, shoot tips look like an inverted boat; mites are microscopic, look for the leaf shape
- 3) THRIPS: leaves curl UPWARD with silvery streaks and tiny black specks of frass on the underside
- 4) All three often occur together, which is why the field looks so badly affected
- 5) Worst in hot dry weather from late Rabi into Kharif-1

C. Immediate actions

- 1) Work out which one it is first: Leaves curling DOWN and looking narrow and shiny = yellow mite (treat with a miticide). Leaves curling UP, thick and leathery, plant stunted, whiteflies flying = virus (no cure, c...)
- 2) If it is the virus, rogue the plants: Bag and remove stunted, upward-curved plants. There is no cure for the virus — removing sources and controlling whitefly protects the rest.
- 3) If it is yellow mite, use a miticide not an insecticide: Abamectin, spiromesifen or wettable sulphur, sprayed on the leaf undersides and growing tips. Ordinary insecticides will not kill mites and will make things worse by killing their...
- 4) Put up yellow and blue sticky traps: Yellow catches whitefly, blue catches thrips. 8-10 per decimal. They also tell you which pest you actually have.
- 5) Stop routine broad-spectrum spraying: Repeated pyrethroid sprays are a leading cause of yellow mite outbreaks in Bangladeshi chilli.

D. Prevention

- 1) Raise seedlings under 40-50 mesh insect-proof net
- 2) Grow tolerant varieties where available
- 3) Yellow and blue sticky traps from the start of the season
- 4) Rogue virus-infected plants weekly
- 5) Do not plant chilli next to old chilli, tomato, brinjal or cotton
- 6) Barrier crops of maize around the plot

E. Fertilizer note

Follow FRG-2018 for chilli. Balanced feeding helps the plant push out new healthy growth after mite damage. Do not add extra nitrogen to virus-infected plants — soft growth attracts more whitefly and the plant will not recover anyway.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Abamectin 1.8% EC (for yellow mite) Abamectin	0.5 ml per litre of water	7	2	7
2	[unverified] Sulphur 80% WP (for yellow mite, lower-ri... Sulphur	2 g per litre of water	10	3	3
3	[AP AP-252] Imidacloprid 20% SL (for whitefly vector ... Imidacloprid [unverified]	0.5 ml per litre of water	12	2	5

Regimen 1 — Timing: Only when yellow mite damage is confirmed (downward leaf curl). Repeat once after 7 days, then rotate. | Water: 3-4 litres of spray solution per decimal | Brands: Vertimec 1.8 EC, Abamec 1.8 EC | PHI note: Do not pick for 7 days after the last spray.

Regimen 2 — Timing: Early mite infestation or as a rotation partner for abamectin. | Water: 3-4 litres of spray solution per decimal | Brands: Thiovit Jet 80 WG, Microthiol Special 80 WP | PHI note: Do not pick for 3 days after the last spray.

Regimen 3 — Timing: Only when whitefly numbers on sticky traps are high. This protects healthy plants; it does not cure virus-infected ones. | Water: 3-4 litres of spray solution per decimal | Brands: Admire 20 SL, Imidor 20 SL, Tido 20 SL | PHI note: Do not pick for 5 days after the last spray.

G. Escalation

Call 16123 if you cannot tell whether it is virus or mite, if more than 20% of plants are stunted, or if curling continues after two miticide sprays.

III.12 Citrus (orange, malta, lemon) — Huanglongbing (citrus greening)

Bangla: লবু জাতীয় ফল (কেমলা, মাল্টা, লবু) — সাইট্রাস গ্রিনিং (হুয়ালংবিং) রোগ। Key: Orange__Huanglongbing_(Citrus_greening).

Pathogen: Candidatus Liberibacter asiaticus, transmitted by Asian citrus psyllid (Diaphorina citri). Severity: high. Pack confidence: medium.

A. Summary

A bacterial disease spread by the citrus psyllid. There is no cure — an infected tree declines and eventually dies. Leaves get blotchy asymmetric yellowing, fruit stays small, green at one end and bitter. Control means removing infected trees,

controlling the psyllid and planting only certified disease-free saplings.

B. Symptoms

- 1) Blotchy, ASYMMETRIC yellowing of leaves — the yellow patches do not match on both sides of the midrib. This is the key sign that separates it from a nutrient deficiency, where yellowing is symmetric.
- 2) Yellow shoots standing out against the green canopy
- 3) Leaves become small, upright and leathery, with corky raised veins
- 4) Fruit is small, lopsided, stays green at the bottom end, and tastes bitter and sour
- 5) Seeds inside are aborted and dark
- 6) Heavy premature fruit drop; branch dieback; the tree declines over 2-5 years

C. Immediate actions

- 1) Report it — this is a regional threat, not just your tree: Call 16123 and inform your Upazila Agriculture Office. Citrus greening destroys whole growing regions once established. Early reporting matters far more than anything you do on yo...
- 2) Do not buy 'greening medicine': There is no cure. Any product sold as a treatment for citrus greening is a waste of money. Trunk injections and foliar nutrients only mask symptoms; they do not remove the bacteri...
- 3) Remove and destroy confirmed infected trees: Cut the tree down and burn or bury it. Spray for psyllids BEFORE removing the tree, otherwise the psyllids on it will simply fly to your healthy trees.
- 4) Control the psyllid on the new flush: The psyllid only breeds on young flush. Time insecticide applications to flush periods, not to a calendar.
- 5) Only plant certified disease-free saplings: Most citrus greening spread in a new area comes in on infected nursery plants. Buy only from a certified nursery, never from a roadside seller.

D. Prevention

- 1) Plant only certified disease-free saplings from a registered nursery
- 2) Remove and destroy infected trees promptly — after spraying for psyllids
- 3) Monitor new flush for psyllid nymphs (white waxy tubules) every 2 weeks
- 4) Do not move citrus planting material between districts
- 5) Remove alternative host plants such as *Murraya* (kamini) from around the orchard
- 6) Keep trees vigorous with balanced nutrition so they tolerate infection longer

E. Fertilizer note

Balanced nutrition, especially zinc, manganese and magnesium, keeps an infected tree productive for longer, but it does not cure the disease. Do not let a nutrition programme become an excuse to keep an infected tree standing — it will keep infecting the neighbourhood.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Imidacloprid 20% SL (for the psyllid vect... Imidacloprid [unverified]	0.5 ml per litre of water	21	3	21

Regimen 1 — Timing: At each new flush, when psyllid nymphs are found. Rotate chemical groups. | Water: Depends on tree size; ensure the new flush is thoroughly covered | Brands: Admire 20 SL, Imidor 20 SL, Tido 20 SL | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 on first suspicion. Do not wait. Report even a single suspicious tree — regional containment depends on early reports.

III.13 Cucumber (and other cucurbits) — Downy mildew

Bangla: শসা (ও অন্যান্য কুমড়া জাতীয় ফসল) — শসার ডাউনি মলিডডি রোগ. Key: Cucumber__Downy_mildew. Pathogen: *Pseudoperonospora cubensis*. Severity: high. Pack confidence: medium.

A. Summary

Angular yellow patches on the upper leaf surface that stop sharply at the leaf veins, with grey-purple fuzz underneath in the morning. In Bangladesh's humid weather it can strip a cucumber field of leaves in 10-14 days.

B. Symptoms

- 1) Angular yellow patches on the upper leaf surface, squared off by the leaf veins — this angular shape is the identifying feature

- 2) Grey to purple-grey downy fungal growth on the UNDERSIDE of the patches, best seen in the early morning
- 3) Patches turn brown and the leaf dries and curls upward
- 4) Starts on the oldest leaves near the base and moves outward along the vine
- 5) Fruit is not directly infected but becomes small and poor-tasting because the leaves have gone
- 6) Explodes in humid, rainy or heavy-dew weather

C. Immediate actions

- 1) Spray immediately — this one moves fast: Downy mildew can defoliate a cucurbit field in under two weeks. A systemic + contact combination (metalaxyl + mancozeb or cymoxanil + mancozeb) at the first angular patch.
- 2) Trellis the vines if you have not: Getting the vines off the ground onto a trellis or bamboo frame dramatically reduces leaf wetness and slows the disease. It also improves fruit quality.
- 3) Remove the worst lower leaves: Strip and bag the heavily affected leaves near the base.
- 4) Stop overhead irrigation: Switch to furrow or drip and irrigate in the morning.
- 5) Keep the protectant going: After the knockdown spray, continue with mancozeb every 7 days while humid weather lasts.

D. Prevention

- 1) Trellis all vining cucurbits — the highest-value structural change
- 2) Wide spacing and good drainage
- 3) Grow tolerant varieties where available
- 4) Remove and destroy crop residue after harvest
- 5) Rotate away from cucurbits for at least one season
- 6) Avoid evening irrigation

E. Fertilizer note

Follow FRG-2018 for cucurbits. Keep potassium adequate — potassium-deficient cucurbits lose leaves to downy mildew much faster. Avoid excess nitrogen, which produces dense humid canopies.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	7	6	5
2	Mancozeb 80% WP + Metalaxyl 8% WP (unverified) Metalaxyl + Mancozeb	2 g per litre of water	10	3	7

[AP AP-374]

Regimen 1 — Timing: Preventive, every 7 days once humid weather starts. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not pick for 5 days after the last spray. Cucumber is eaten raw — respect this strictly.

Regimen 2 — Timing: Curative, at the first angular yellow patches. Maximum 3 per season, always alternated with mancozeb. | Water: 3-4 litres of spray solution per decimal | Brands: Ridomil Gold MZ 68 WG, Galben M, Krilaxyl Gold | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if leaves are drying across the field, if the disease continues after two sprays, or if you are unsure whether it is downy or powdery mildew — they need different products.

III.14 Cucumber (and other cucurbits) — Powdery mildew

Bangla: শসা (ও অন্যান্য কুমড়া জাতীয় ফসল) — শসার পাউডারি মলিডডি রোগ. Key: Cucumber__Powdery_mildew. Pathogen: Podosphaera xanthii / Erysiphe cichoracearum. Severity: medium. Pack confidence: medium.

A. Summary

White powdery patches on the top of the leaves that look like someone dusted them with flour, and can be wiped off with a finger. Unlike downy mildew it likes dry weather, and it is easily controlled with sulphur if you catch it early.

B. Symptoms

- 1) White to greyish powdery patches on the UPPER leaf surface and on stems — wipes off with a finger
- 2) Patches spread and merge until the whole leaf looks white
- 3) Leaves turn yellow, then brown, dry and die
- 4) Fruit is not directly infected but is small and sunburnt once the leaves go

- 5) Unlike downy mildew, it does NOT need leaf wetness — it thrives in dry warm weather with high humidity at night
- 6) Older leaves and shaded lower leaves are affected first

C. Immediate actions

- 1) Confirm it is powdery, not downy: Powdery mildew is WHITE POWDER on the TOP of the leaf and wipes off. Downy mildew is ANGULAR YELLOW patches on top with GREY FUZZ underneath. Different products — check before buy...
- 2) Spray wettable sulphur: 2 g per litre, covering both leaf surfaces. Cheap, effective and safe if the temperature is below 32°C.
- 3) Do not use sulphur in hot weather: Above 32°C sulphur scorches cucurbit leaves badly. In hot weather use hexaconazole or difenoconazole instead.
- 4) Remove the worst lower leaves: Strip heavily coated old leaves and take them out of the field.
- 5) Improve air movement: Trellis the vines and thin dense growth.

D. Prevention

- 1) Trellis vines and space plants widely
- 2) Grow tolerant varieties where available
- 3) Remove old lower leaves regularly
- 4) Do not grow cucurbits in heavy shade
- 5) Destroy crop residue after harvest
- 6) Rotate away from cucurbits for a season

E. Fertilizer note

Follow FRG-2018 for cucurbits. Avoid excess nitrogen, which produces soft dense growth. Adequate potassium and silicon help the leaf resist infection.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Sulphur 80% WP / WG (wetable sulphur) Sulphur	2 g per litre of water	10	4	3
2	[AP AP-252] Hexaconazole 5% EC Hexaconazole	2 ml per litre of water	12	2	7

Regimen 1 — Timing: At the first white patches; repeat every 10 days while conditions persist. | Water: 3-4 litres of spray solution per decimal | Brands: Thiovit Jet 80 WG, Microthiol Special 80 WP, Sulcox 80 WP | PHI note: Do not pick for 3 days after the last spray.

Regimen 2 — Timing: When it is too hot for sulphur, or when powdery mildew is well established. Maximum 2 per season. | Water: 3-4 litres of spray solution per decimal | Brands: Contaf 5 EC, Hexa 5 EC | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if the whole canopy is whitening, if leaves are dying rapidly, or if you cannot tell powdery from downy mildew.

III.15 Grape — Black rot

Bangla: আঙুর — আঙুরের ব্ল্যাক রট রোগ. Key: Grape__Black_rot. Pathogen: Guignardia bidwellii. Severity: medium. Pack confidence: high.

A. Summary

A disease of grapevines, which are not a commercial crop in Bangladesh. Included for model completeness. If AgroScan returns this for a Bangladeshi field photo, treat it as a probable misclassification and re-photograph.

B. Symptoms

- 1) Small reddish-brown circular leaf spots with dark borders and black dots inside
- 2) Fruit shrivels into hard black mummies
- 3) Lesions on shoots and tendrils

C. Immediate actions

- 1) Check whether this is really grape: Grape is not a commercial crop in Bangladesh. Retake the photo of a clearer leaf and rescan.
- 2) Call 16123 if unsure of the crop: Take a sample leaf to your Upazila Agriculture Office.
- 3) Do not spray on this result: Do not treat a crop you are not growing.

D. Prevention

1) If you do grow grape experimentally: remove mummified fruit, prune for airflow, and contact BARI Horticulture Research Centre for guidance

E. Fertilizer note

Not applicable at scale in Bangladesh. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 to identify your crop.

III.16 Grape — Esca (black measles)

Bangla: আঙুর — আঙুরের এসকা (ব্ল্যাক মজিলস) রোগ। Key: Grape__Esca_(Black_Measles). Pathogen: Phaeomoniella chlamydospora and associated fungi. Severity: medium. Pack confidence: high.

A. Summary

A trunk disease of grapevines, not relevant to Bangladeshi farming. Included for model completeness. Treat a Bangladeshi prediction as a probable misclassification.

B. Symptoms

- 1) Tiger-stripe pattern of yellow or red between the veins on leaves
- 2) Small dark spots on berries
- 3) Sudden collapse of whole vines in hot weather

C. Immediate actions

- 1) Check whether this is really grape: Retake the photo of a clearer leaf and rescan.
- 2) Call 16123 if unsure: Take a sample leaf to your Upazila Agriculture Office.

D. Prevention

- 1) Not applicable in Bangladesh

E. Fertilizer note

Not applicable. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 to identify your crop.

III.17 Grape — Isariopsis leaf spot / leaf blight

Bangla: আঙুর — আঙুরের পাতার দাগ রোগ। Key: Grape__Leaf_blight_(Isariopsis_Leaf_Spot). Pathogen: Pseudocercospora vitis. Severity: low. Pack confidence: high.

A. Summary

A leaf-spot disease of grapevines, not relevant to Bangladeshi farming. Included for model completeness.

B. Symptoms

- 1) Irregular dark brown blotches on leaves
- 2) Premature defoliation in wet weather

C. Immediate actions

- 1) Check whether this is really grape: Retake the photo of a clearer leaf and rescan.
- 2) Call 16123 if unsure: Take a sample leaf to your Upazila Agriculture Office.

D. Prevention

- 1) Not applicable in Bangladesh

E. Fertilizer note

Not applicable. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 to identify your crop.

III.18 Maize — Common rust

Bangla: ভুট্টা — ভুট্টার সাধারণ মরিচা রোগ. Key: Corn_(maize)___Common_rust_. Pathogen: Puccinia sorghi. Severity: medium. Pack confidence: medium.

A. Summary

Small cinnamon-brown powdery pustules on both sides of the leaf that rub off orange onto your fingers. It likes cool humid weather and is common in Bangladesh's Rabi maize. Most hybrid maize grown here tolerates it well, so spraying is only worth it if it starts before flowering.

B. Symptoms

- 1) Small oval to elongated cinnamon-brown pustules scattered on BOTH upper and lower leaf surfaces
- 2) Pustules break open and release orange-brown powder that rubs off on your hand
- 3) Later in the season pustules turn brownish-black
- 4) In heavy attacks leaves dry prematurely and stalks weaken
- 5) Favoured by cool (16-25°C) humid weather with heavy dew

C. Immediate actions

- 1) Judge the growth stage first: If the crop has already passed grain filling, rust will not cost you much and spraying is not worth the money. If it is before or at tasselling, act.
- 2) Count affected leaves: Spray only if rust is on the ear leaf or the two leaves above it, and more than about 5-10% of that leaf area is covered.
- 3) Spray a triazole or strobilurin: Propiconazole, tebuconazole or azoxystrobin at label rate, covering the upper leaves that feed the cob.
- 4) Do not add nitrogen: Extra urea at this stage will not help and produces soft tissue that rusts more.

D. Prevention

- 1) Grow rust-tolerant hybrids — most commercial hybrids sold in Bangladesh have useful tolerance
- 2) Do not plant maize too densely; wider rows dry faster
- 3) Rotate and destroy infected residue after harvest
- 4) Plant at the recommended time so grain filling finishes before peak rust weather
- 5) Apply balanced potash, which improves stalk strength and disease tolerance

E. Fertilizer note

Follow FRG-2018 for maize. Ensure the full potash dose — potassium improves both stalk strength and rust tolerance. Do not apply late extra nitrogen.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Propiconazole 25% EC Propiconazole [unverified]	1 ml per litre of water (about 10 ml per 10 litre sprayer load)	15	2	30

Regimen 1 — Timing: At tasselling to early silking if rust reaches the ear leaf. One spray is usually enough. | Water: 4-5 litres of spray solution per decimal on a tall crop | Brands: Tilt 250 EC, Propicon 250 EC | PHI note: Do not harvest for 30 days after the last spray. Do not feed sprayed green maize to cattle inside this period.

G. Escalation

Call 16123 if rust covers the ear leaf before tasselling or if leaves are drying prematurely across the field.

III.19 Maize — Gray leaf spot

Bangla: ভুট্টা — ভুট্টার ধূসর পাতার দাগ রোগ. Key: Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot. Pathogen: Cercospora zeae-maydis / C. zeina. Severity: medium. Pack confidence: low.

A. Summary

Narrow rectangular grey to tan lesions that run in straight lines between the leaf veins, like matchsticks laid on the leaf. It builds up where maize follows maize with the residue left on the surface.

B. Symptoms

- 1) Narrow, rectangular lesions with straight parallel sides, bounded by the leaf veins
- 2) Lesions start grey-green and turn grey to tan, 2-6 cm long and 2-4 mm wide
- 3) Held up to the light, the lesion edges are sharp and straight — this distinguishes it from northern leaf blight's rounded cigar shape
- 4) Starts on lower leaves and moves up; lesions merge and kill whole leaves
- 5) Worst in continuous maize with surface residue, in warm humid weather with long dew

C. Immediate actions

- 1) Confirm it is gray leaf spot: Hold a leaf up to the sunlight. Gray leaf spot lesions are rectangular with straight edges stopped by the veins. Northern leaf blight lesions are cigar-shaped with rounded ends th...
- 2) Check the position on the plant: Spray only if lesions have reached the leaf below the cob before or at tasselling. Below that, the yield effect is small.
- 3) Spray a strobilurin + triazole: Azoxystrobin + difenoconazole or propiconazole at label rate, targeting the upper canopy at tasselling.
- 4) Fix the residue problem: The real cure is next season: plough in the residue or rotate out of maize. Surface maize trash is where this fungus lives.

D. Prevention

- 1) Rotate out of maize for at least one season — the single most effective measure
- 2) Plough residue under rather than leaving it on the surface
- 3) Grow tolerant hybrids
- 4) Avoid dense planting; keep rows open for air movement
- 5) Apply balanced fertiliser at FRG rates

E. Fertilizer note

Follow FRG-2018 for maize with the full potash dose. Balanced nutrition helps the plant hold its upper leaves through grain filling.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Azoxystrobin 20% + Difenoconazole 12.5% SC Azoxystrobin + Difenoconazole [AP AP-2312]	1 ml per litre of water	15	2	30

Regimen 1 — Timing: At tasselling, once, if lesions have reached the leaf below the cob. | Water: 4-5 litres of spray solution per decimal | Brands: Amistar Top | PHI note: Do not harvest for 30 days after the last spray.

G. Escalation

Call 16123 if lesions reach above the cob before grain filling or if you cannot distinguish this from northern leaf blight.

III.20 Maize — Northern corn leaf blight

Bangla: ভুট্টা — ভুট্টার পাতা বলসানো রোগ। Key: Corn_(maize)___Northern_Leaf_Blight. Pathogen: Exserohilum turcicum (Setosphaeria turcica). Severity: high. Pack confidence: medium.

A. Summary

Long grey-green cigar-shaped lesions on the leaves, starting on the lower leaves and moving up. If these reach the leaves above the cob before grain filling, yield drops sharply. Cool wet weather with long dew periods drives it.

B. Symptoms

- 1) Long (2-15 cm) elliptical, cigar-shaped grey-green to tan lesions running along the leaf
- 2) Lesions have smooth edges and are much bigger than rust pustules or gray leaf spot
- 3) Starts on the lowest leaves and moves upward
- 4) In severe attacks lesions merge and whole leaves die, giving a scorched look
- 5) Dark olive fungal growth visible on lesions in humid weather
- 6) Favoured by 18-27°C with 6+ hours of leaf wetness

C. Immediate actions

- 1) Check how high the lesions have climbed: If lesions are still below the cob, you have time. If they are on the ear leaf or above before grain filling, spray now — this is where the yield loss happens.

2) Spray a triazole or strobilurin: Propiconazole, tebuconazole or azoxystrobin at label rate, targeting the upper canopy. One well-timed spray at tasselling is usually worth more than two late ones.

3) Do not add extra nitrogen: Soft nitrogen-rich tissue blights faster. Stick to the FRG rate.

4) Plan crop rotation now: The fungus survives on maize residue. If you plant maize after maize on the same land, this disease will get worse every year. Decide the rotation before the next season.

D. Prevention

- 1) Grow resistant or tolerant hybrids — the most effective and cheapest control
- 2) Rotate maize with rice, wheat, potato or legumes for at least one season
- 3) Plough in or remove maize residue after harvest; the fungus overwinters there
- 4) Avoid very dense planting; keep rows wide enough for air movement
- 5) Plant on time so the crop passes grain filling before the wettest period
- 6) Apply balanced potash

E. Fertilizer note

Follow FRG-2018 for maize. Apply the full potash dose. Avoid heavy late nitrogen, which delays maturity and extends the window in which the disease can damage the crop.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Propiconazole 25% EC	1 ml per litre of water	15	2	30
2	Propiconazole 20% + Difenoconazole 12.5% SC (unverified) Azoxystrobin + Difenoconazole	1 ml per litre of water	15	2	30

[AP AP-2312]

Regimen 1 — Timing: At tasselling, or earlier if lesions reach the ear leaf before then. | Water: 4-5 litres of spray solution per decimal | Brands: Tilt 250 EC, Propicon 250 EC | PHI note: Do not harvest for 30 days after the last spray; do not feed sprayed green maize to livestock inside this period.

Regimen 2 — Timing: At tasselling. Maximum 2 per season; rotate with a triazole so strobilurin resistance does not build. | Water: 4-5 litres of spray solution per decimal | Brands: Amistar Top | PHI note: Do not harvest for 30 days after the last spray.

G. Escalation

Call 16123 if lesions reach the ear leaf before tasselling, if more than 30% of leaf area is blighted, or if the same field gets it every year despite rotation.

III.21 Mango — Mango anthracnose

Bangla: আম — আমের অ্যানথ্রাকনোজ রোগ. Key: Mango__Anthracnose. Pathogen: Colletotrichum gloeosporioides. Severity: high. Pack confidence: high.

A. Summary

Black sunken spots on leaves, blackened flower panicles and black rotten patches on the fruit — often appearing only after the fruit is picked and ripening. In Bangladesh's mango belt the flowering-time sprays decide the crop, because infection happens at flowering and stays hidden until ripening.

B. Symptoms

- 1) Small dark brown to black irregular spots on leaves that later dry and drop out leaving holes
- 2) Blackening and drying of flower panicles — this is where the crop is lost
- 3) Black sunken spots on young fruit, causing fruit drop
- 4) On ripening fruit: black sunken lesions, often in a tear-stain pattern running down the fruit, that spread and rot the flesh
- 5) Pink spore masses on lesions in wet weather
- 6) Worst when it rains or there is heavy dew during flowering (February-March)

C. Immediate actions

1) Spray at the right growth stages, not randomly: The three sprays that matter are: (1) at panicle emergence, (2) at full bloom or just after, and (3) at pea-stage fruit. Sprays outside these windows do very little.

2) Do not spray during full bloom if bees are working: Mango needs insect pollination. Spray early morning or late evening, or shift the spray to just after petal fall.

3) Prune out dead twigs and mummified fruit: The fungus overwinters in dead twigs, old panicles and dried fruit left in the tree. Remove and burn them before flowering.

4) Open up the canopy: Prune to let light and air into the centre of the tree. A dense mango canopy stays wet for hours after dew and that is what the fungus needs.

5) Handle fruit carefully at harvest: Harvest with 1-2 cm of stalk attached, avoid sap burn, and do not bruise the fruit. Latent infections become visible rot during ripening and transport.

D. Prevention

- 1) Prune annually after harvest to open the canopy
- 2) Remove and burn dead twigs, old panicles and mummified fruit before flowering
- 3) Keep the orchard floor clean of fallen leaves and fruit
- 4) Do not plant too densely; maintain proper tree spacing
- 5) Harvest at the correct maturity with stalk attached; de-sap carefully
- 6) Hot water treatment of harvested fruit at 52-55°C for 5 minutes greatly reduces post-harvest anthracnose

E. Fertilizer note

Apply the annual dose after harvest, not during flowering. Avoid heavy nitrogen before flowering — it produces vegetative flushes instead of panicles and makes anthracnose worse. Potassium and calcium improve fruit skin quality and shelf life.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	15	4	15
2	Mancozeb Azoxystrobin 20% + Cyproconazole 8% SC (unverified) Azoxystrobin + Cyproconazole	1 ml per litre of water (label field rate recorded as 1200 ml per hec...	15	2	21
3	AP AP-24311 Propiconazole 25% EC	1 ml per litre of water	15	2	21

Regimen 1 — Timing: Preventive: at panicle emergence and again at pea-stage fruit. | Water: A mature mango tree needs roughly 40-100 litres of spray solution depending on ... | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not harvest for 15 days after the last spray. Mango for export has strict residue limits — check with your buyer.

Regimen 2 — Timing: At panicle emergence and after petal fall. Maximum 2 per season; alternate with mancozeb. | Water: Volume depends on tree size; ensure full canopy coverage | Brands: Karishma 28 SC | PHI note: Do not harvest for 21 days after the last spray.

Regimen 3 — Timing: At flowering and pea stage. Rotate with mancozeb and the strobilurin mix. | Water: Volume depends on tree size | Brands: Tilt 250 EC, Propicon 250 EC | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 if panicles are blackening across the orchard, if there is heavy fruit drop at pea stage, or if fruit is rotting during ripening despite field sprays.

III.22 Mango — Mango powdery mildew

Bangla: আম — আমরে পাউডার মিলিডডি রোগ. Key: Mango__Powdery_mildew. Pathogen: Erysiphe quercicola (Oidium mangiferae).

Severity: high. Pack confidence: medium.

A. Summary

A white powdery coating on the flower panicles, young leaves and small fruit. It hits at flowering time, dries the panicle and causes heavy fruit drop. Dry weather with cool nights and morning dew is exactly what it likes — the opposite of anthracnose.

B. Symptoms

- 1) White or greyish powdery growth on panicles, young leaves and small fruit — wipes off with a finger
- 2) Affected panicles dry up and turn brown without setting fruit
- 3) Young fruit drops in large numbers; retained fruit is deformed and russeted
- 4) Young leaves curl and become distorted
- 5) Worst in dry weather with cool nights (10-20°C) and heavy morning dew during flowering

C. Immediate actions

1) Act at the first white patch on a panicle: Powdery mildew moves through an orchard in days at flowering. One spray on the day you first see it is worth three sprays a week later.

2) Use sulphur or a triazole: Wettable sulphur at 2 g/L is cheap and effective, or hexaconazole/difenoconazole at label rate. Cover the panicles thoroughly.

3) Do not use sulphur in hot weather: Above about 32°C sulphur scorches young leaves and fruit. If it is hot, use a triazole instead.

4) Protect bees: Spray very early morning or late evening. Mango pollination depends on flies and bees visiting the panicles.

5) Repeat at 10-15 days if conditions persist: Two to three sprays across the flowering period are usually needed in a bad year.

D. Prevention

- 1) Prune to open the canopy so dew dries early
- 2) Remove and destroy affected panicles where practical
- 3) Avoid excess nitrogen before flowering
- 4) Keep the orchard floor clean
- 5) Plan the flowering-stage spray schedule before the panicles emerge, not after symptoms appear

E. Fertilizer note

Apply the annual dose after harvest. Avoid nitrogen before flowering. Adequate potassium supports better fruit set and retention after a mildew attack.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Sulphur 80% WP / WG (wetttable sulphur) Sulphur	2 g per litre of water	12	3	7
2	[AP AP-252] Hexaconazole 5% EC	2 ml per litre of water	15	2	21

Regimen 1 — Timing: At the first sign of white powder on panicles; repeat after 10-15 days if conditions persist. | Water: Depends on tree size; ensure panicles are fully wetted | Brands: Thiovit Jet 80 WG, Microthiol Special 80 WP, Sulcox 80 WP | PHI note: Do not harvest for 7 days after the last spray.

Regimen 2 — Timing: At panicle emergence as a preventive, or at first symptoms. Safe to use in warm weather when sulphur is not. | Water: Depends on tree size | Brands: Contaf 5 EC, Hexa 5 EC | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 if panicles across the orchard are drying without setting fruit, or if fruit drop is heavy after flowering.

III.23 Onion — Purple blotch

Bangla: পঁয়োজ — পঁয়োজের পার্পল ব্লচ রোগ. Key: Onion__Purple_blotch. Pathogen: Alternaria porri. Severity: high. Pack confidence: high.

A. Summary

Small white sunken spots on the leaves that grow into large purplish-brown zoned patches, then the leaf collapses at that point. It is the main foliar disease of onion in Bangladesh and it also rots the bulb neck in storage.

B. Symptoms

- 1) Small white or greyish sunken spots on the leaf, often with a purple centre as they enlarge
- 2) Spots grow into large zoned (ringed) purple-brown patches with a yellow halo above and below
- 3) The leaf breaks over and collapses at the lesion
- 4) Bulb size drops sharply because the leaves die early
- 5) Neck rot in storage: the neck becomes water-soaked, then yellow-red, then rots
- 6) Worst in warm humid weather, heavy dew and after irrigation over the top

C. Immediate actions

- 1) Spray with a sticker — this is the key detail: Onion leaves are upright and waxy, so spray runs straight off. Always add a wetting agent or sticker at 1 ml/L. Farmers who skip this often conclude the fungicide 'did not work'.
- 2) Use mancozeb or a difenoconazole product: Mancozeb at 2-2.5 g/L as the protectant backbone, or difenoconazole at 0.5 ml/L where the disease is established.
- 3) Stop overhead irrigation: Switch to furrow irrigation. Keeping the leaves dry matters as much as the spray.
- 4) Remove badly affected leaves in small plots: Cut off collapsed leaves and take them out of the field.
- 5) Cure the bulbs properly at harvest: After lifting, cure the bulbs in shade with good airflow for 2-3 weeks until the necks are tight and dry. This is what prevents storage neck rot.

D. Prevention

- 1) Use certified, disease-free sets or seed

- 2) Treat seed before sowing
- 3) Rotate away from onion, garlic and other alliums for 2-3 years
- 4) Plant on raised beds with good drainage
- 5) Do not plant too densely
- 6) Use furrow irrigation, never overhead, and irrigate in the morning

E. Fertilizer note

Follow FRG-2018 for onion. Adequate potassium and sulphur improve bulb quality and storage life. Avoid excess nitrogen, which produces soft thick necks that do not cure properly and rot in storage.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP + wetting agent Mancozeb	2-2.5 g per litre of water plus 1 ml/L wetting agent	10	5	10
2	Unverified Difenoconazole 25% EC Difenoconazole	0.5 ml per litre of water plus a wetting agent	12	2	14

[AP AP-384]

Regimen 1 — Timing: Preventive, every 10-12 days from 30-35 days after planting. | Water: 2-3 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not harvest for 10 days after the last spray.

Regimen 2 — Timing: Curative, when blotches are already forming. Maximum 2 per season, alternated with mancozeb. | Water: 2-3 litres of spray solution per decimal | Brands: Score 250 EC, Difen 250 EC | PHI note: Do not harvest for 14 days after the last spray.

G. Escalation

Call 16123 if leaves are collapsing across the field before bulbing is complete, or if stored onions are developing neck rot.

III.24 Peach — Bacterial spot

Bangla: পিচ ফল — পিচ ফলের ব্যাকটেরিয়াজনিত দাগ রোগ. Key: Peach__Bacterial_spot. Pathogen: Xanthomonas arboricola pv. pruni.

Severity: medium. Pack confidence: high.

A. Summary

A disease of peach orchards in temperate climates. Peach is not grown commercially in Bangladesh. Included for model completeness only.

B. Symptoms

- 1) Small angular dark spots on leaves that fall out leaving shot holes
- 2) Sunken dark spots and cracking on fruit
- 3) Cankers on twigs

C. Immediate actions

- 1) Check whether this is really peach: Peach is not grown commercially in Bangladesh. Retake the photo of a clearer leaf and rescan.
- 2) Call 16123 if unsure: Take a sample leaf to your Upazila Agriculture Office.
- 3) Do not spray on this result: Do not treat a crop you are not growing.

D. Prevention

- 1) Not applicable in Bangladesh

E. Fertilizer note

Not applicable. Identify the actual crop first.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 to identify your crop.

III.25 Potato — Early blight

Bangla: আলু — আলুর আগাম ধ্বসা রোগ. Key: Potato__Early_blight. Pathogen: Alternaria solani. Severity: medium. Pack confidence: medium.

A. Summary

Early blight makes brown spots with dark rings inside them, like a target board, starting on the older lower leaves. It is slower and less deadly than late blight but it steals yield quietly, and it hits hardest where the crop is short of nitrogen or under stress.

B. Symptoms

- 1) Brown to dark brown round spots with concentric rings inside, like a target
- 2) Starts on the oldest, lowest leaves and works upward
- 3) A yellow halo often surrounds the spot
- 4) Leaves yellow, dry and drop; the plant looks bare from the bottom up
- 5) Dark, sunken, dry lesions on tubers with a raised rim, rot is dry and firm not slimy
- 6) Worse in warm weather with alternating wet and dry spells, and on nitrogen-starved plants

C. Immediate actions

- 1) Tell it apart from late blight first: Early blight spots have rings inside and start on the OLD lower leaves; the plant does not collapse. Late blight starts anywhere, has a white downy edge in the morning and collapses...
- 2) Feed the crop: Early blight attacks weak, hungry plants. If leaves are pale, give a nitrogen top-dress at the FRG rate. This is the opposite of late blight advice.
- 3) Remove the lowest infected leaves: Pick off and remove badly spotted bottom leaves in small plots. This slows the climb up the plant.
- 4) Spray a protectant: Mancozeb or chlorothalonil at 7-10 day intervals. If it is already well established, use a difenoconazole or azoxystrobin product once, then go back to protectants.
- 5) Keep the soil evenly moist: Wet-dry stress cycles make early blight worse. Irrigate regularly rather than letting the crop dry out and then flooding it.

D. Prevention

- 1) Rotate with non-solanaceous crops (rice, maize, mustard) for at least 2 years
- 2) Use certified seed tubers and destroy infected crop residue after harvest
- 3) Maintain adequate nitrogen and potassium — hungry crops get early blight badly
- 4) Avoid overhead irrigation late in the day
- 5) Space plants so the canopy dries; earth up ridges well
- 6) Harvest at full maturity and cure tubers before storage so lesions do not develop in store

E. Fertilizer note

Unlike late blight, early blight is a disease of hungry plants. Make sure the full FRG-2018 nitrogen dose is applied on schedule and keep potassium adequate. Pale, prematurely yellowing plants are an invitation to *Alternaria*.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP Mancozeb	2-2.5 g per litre of water (about 20-25 g per 10 litre sprayer load)	7	6	7
2	[unverified] Difenoconazole 25% EC Difenoconazole [AP AP-384]	0.5 ml per litre of water (about 5 ml per 10 litre sprayer load)	12	2	14

Regimen 1 — Timing: Preventive, from 30-35 days after planting, every 7-10 days. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not harvest for 7 days after the last spray.

Regimen 2 — Timing: Curative — use when spots are already spreading. Maximum 2 sprays per season, alternated with mancozeb. | Water: 3-4 litres of spray solution per decimal | Brands: Score 250 EC, Difen 250 EC | PHI note: Do not harvest for 14 days after the last spray.

G. Escalation

Call 16123 if more than 30% of leaves are affected, if you are not sure whether it is early or late blight, or if the disease is climbing quickly to the top of the plants.

III.26 Potato — Late blight

Bangla: আলু — আলুর নাবিধ্বসা রোগ. Key: Potato__Late_blight. Pathogen: *Phytophthora infestans*. Severity: high. Pack confidence: high.

A. Summary

Late blight is the most destructive potato disease in Bangladesh and can destroy a whole field in 7-10 days in foggy December-January weather. It is a race against time: preventive spraying before the fog sets in saves the crop, spraying after the field turns black does not.

B. Symptoms

- 1) Water-soaked dark green to blackish-brown patches on leaves, usually starting at the leaf tip or edge
- 2) In the early morning a white downy fungal ring on the underside of the lesion, right at the edge of the dead tissue
- 3) Lesions spread fast in fog; whole leaves collapse and the field smells of rotting potato
- 4) Dark brown to black lesions on stems and petioles that girdle and break them
- 5) Firm reddish-brown rot spreading irregularly from the tuber skin inwards; infected tubers rot in storage
- 6) Explodes when night temperature is 10-20°C with fog or drizzle and long leaf wetness

C. Immediate actions

- 1) Spray today, not tomorrow: Late blight doubles roughly every 2 days in fog. A spray today is worth three sprays next week. If you see the first water-soaked patches, spray the same day.
- 2) Switch to a systemic + contact mix: Once lesions are visible, a contact fungicide alone is not enough. Use a systemic combination such as metalaxyl + mancozeb or cymoxanil + mancozeb, then go back to contact product...
- 3) Remove and bury the worst plants: Cut out badly infected plants and bury them at least 30 cm deep away from the field or burn them. Do not leave them on the bund — they keep producing spores.
- 4) Stop overhead irrigation: Do not irrigate by flooding or sprinkling while the disease is active. Wet leaves are what the disease needs. Irrigate in the morning, never in the evening.
- 5) Earth up the ridges: Pull soil up over the tubers so spores washing down from the leaves cannot reach them. This is what prevents tuber rot in storage.
- 6) Plan the harvest: If more than half the canopy is dead, cut and remove the haulm (tops), wait 10-14 days for the skin to set, then harvest on a dry day. Do not harvest wet — wet infected tubers rot...

D. Prevention

- 1) Plant only certified, blight-free seed tubers from BADC or a licensed dealer
- 2) Plant early (mid-October to early November) so the crop bulks before peak fog
- 3) Use tolerant varieties where available; Diamant and Cardinal are widely grown but are not resistant
- 4) Keep wide ridges and good drainage so the canopy dries quickly
- 5) Do not plant potato next to a tomato field — the same pathogen moves between them
- 6) Rotate away from potato and tomato for at least 2 years on badly affected land

E. Fertilizer note

Avoid excess nitrogen — soft, lush foliage is far more susceptible and stays wet longer. Apply the full potash dose from FRG-2018; potassium thickens cell walls and reduces both foliar and tuber infection. Do not top-dress urea once blight has started.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP Mancozeb	2-2.5 g per litre of water (about 20-25 g per 10 litre sprayer load)	7	6	7
2	[AP AP-374] Metalaxyl 8% + Mancozeb 64% WP Metalaxyl (or Benalaxyl) + Mancozeb	2 g per litre of water (about 20 g per 10 litre sprayer load); label ...	10	3	14
3	[AP AP-374] Cymoxanil 8% + Mancozeb 64% WP Cymoxanil + Mancozeb [unverified]	2-2.5 g per litre of water (about 20-25 g per 10 litre sprayer load)	10	3	14

Regimen 1 — Timing: PREVENTIVE. Start at 30-35 days after planting or as soon as fog begins, and repeat every 7-10 days. This is the backbone spray of the season. | Water: 3-4 litres of spray solution per decimal at full canopy | Brands: Dithane M-45, Indofil M-45, Manco 80 WP, Emcozeb 80 WP | PHI note: Do not harvest for 7 days after the last spray.

Regimen 2 — Timing: CURATIVE/knockdown. Use at the first visible lesions, or as the first spray when the disease is already in the neighbourhood. Use a maximum of 2-3 times per se... | Water: 3-4 litres of spray solution per decimal | Brands: Galben M, Ridomil Gold MZ 68 WG, Krilaxyl Gold | PHI note: Do not harvest for 14 days after the last spray.

Regimen 3 — Timing: At first lesions, or as an alternate knockdown spray in a heavy blight year. | Water: 3-4 litres of spray solution per decimal | Brands: Curzate M8, Melody Duo, Secure 60 WG | PHI note: Do not harvest for 14 days after the last spray.

G. Escalation

Call 16123 immediately if lesions are spreading despite two sprays, if more than 20% of the canopy is affected, or if blight has appeared in several fields in your area. Late blight is a village-level emergency, not an individual problem.

III.27 Rice — Bacterial leaf blight (BLB)

Bangla: ধান — ধানের ব্যাকটেরিয়াজনিত পাতা পোড়া রোগ। Key: Rice__Bacterial_leaf_blight. Pathogen: Xanthomonas oryzae pv. oryzae. Severity: high. Pack confidence: high.

A. Summary

BLB is a bacterial disease, so ordinary fungicides do nothing against it. Leaves dry from the tip and edges inwards in wavy yellow streaks. It spreads through flood water, wind-driven rain and cut leaf tips, and it is worst in Aman after storms.

B. Symptoms

- 1) Yellow to straw-coloured wavy streaks starting at the leaf tip or edge and moving down
- 2) In early morning, milky or yellowish bacterial ooze beads on fresh lesions — press a cut leaf end in clear water and you will see a cloudy stream
- 3) Whole leaves dry to straw colour and the field looks scorched
- 4) Kresk phase: young transplanted seedlings wilt and die completely within days
- 5) Follows storms, flooding, and fields given heavy urea

C. Immediate actions

- 1) Do NOT spray a fungicide: Fungicides do not work on bacteria. Spraying one wastes money and delays the actions that actually help. Many farmers lose the crop this way.
- 2) Stop all nitrogen immediately: Hold back every remaining urea top-dress. Nitrogen makes BLB spread dramatically faster. This is the single most important step.
- 3) Drain the field: Drain standing water and keep the field just moist for 3-4 days. Standing water carries the bacteria from plant to plant.
- 4) Apply potash: Top-dress 5-7 kg MoP per bigha (33 decimal). Potash noticeably slows BLB.
- 5) Stop moving through the field: The bacteria spread on wet clothes, hands and tools. Do not walk through a wet infected field, and never move from an infected field into a clean one without washing.
- 6) Do not clip seedling tips: Clipping leaf tips at transplanting is a main entry route for the bacteria. Stop the practice in BLB-prone areas.

D. Prevention

- 1) Grow resistant varieties — BRRI dhan51, BRRI dhan52, BRRI dhan62, BRRI dhan75 and newer BLB-resistant BRRI releases
- 2) Use certified, disease-free seed from BADC or a licensed dealer
- 3) Split nitrogen into 3 doses and never exceed the FRG rate
- 4) Do not transplant into flooded fields immediately after a storm
- 5) Keep the field free of weedy rice and grass weeds, which host the bacteria
- 6) Burn or deep-bury infected straw and stubble; do not feed it to cattle that will return manure to the same field

E. Fertilizer note

Cut nitrogen, raise potash. Do not apply any urea while the disease is active. Give the full MoP dose and consider a foliar potash spray. Resume normal nitrogen only after new leaves come out clean.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Copper oxychloride 50% WP Copper oxychloride [AP AP-354]	3 g per litre of water (about 30 g per 10 litre sprayer load)	10	2	15

Regimen 1 — Timing: As soon as streaks are confirmed, and again after 10 days if the disease is still moving. Do not spray during flowering. | Water: 4-5 litres of spray solution per decimal | Brands: Cupravit 50 WP, Champion 77 WP, Blue Copper | PHI note: Do not harvest for 15 days after the last spray. Copper accumulates in soil — do not use it season after season on the ...

G. Escalation

Call 16123 immediately if seedlings are wilting and dying (kresk), if more than 20% of the field is scorched, or if the disease appeared after flood water entered the field. BLB outbreaks after floods need Upazila-level coordination.

III.28 Rice — Brown spot

Bangla: ধান — ধানের বাদামি দাগ রোগ। Key: Rice__Brown_spot. Pathogen: Bipolaris oryzae (Cochliobolus miyabeanus). Severity: medium. Pack confidence: high.

A. Summary

Brown spot is mainly a hunger disease — it shows up on poor, sandy or drought-stressed soil where the crop is short of nitrogen, potash or silicon. Oval brown spots cover the leaves and the grain becomes discoloured and light. Fixing the soil matters more than spraying.

B. Symptoms

- 1) Round to oval brown spots on leaves, like sesame seeds, with a grey centre and dark brown edge
- 2) Unlike blast, the spots are blunt and oval — not pointed and diamond-shaped
- 3) Many small spots on the seedling leaves in the nursery, sometimes killing seedlings
- 4) Brown or black spots on the glumes; grain is shrivelled and discoloured (dirty rice)
- 5) Worst on droughty, sandy or unfertilised land and in fields short of potash

C. Immediate actions

- 1) Check the water first: Brown spot almost always follows water stress. Keep 3-5 cm of standing water in the field. If the field has been dry, irrigate immediately — this alone often stops it.
- 2) Feed the crop: Top-dress urea if the leaves are pale yellow and the crop is clearly underfed. Unlike blast, brown spot gets BETTER with adequate nitrogen, not worse.
- 3) Apply potash: Give 5-7 kg MoP per bigha (33 decimal). Potash deficiency is the single most common cause of brown spot in Bangladesh.
- 4) Spray only if severe: If more than 10% of leaf area is spotted or the flag leaf is affected, spray a propiconazole or mancozeb product. Light infection on a well-fed crop does not need spraying.

D. Prevention

- 1) Treat seed before sowing — this is the cheapest and most effective step
- 2) Do not save seed from a field that had discoloured, spotted grain
- 3) Apply the full FRG dose of potash and, on old rice land, add zinc sulphate
- 4) Incorporate rice straw or husk ash to restore silicon on long-cropped land
- 5) Level the field so water does not drain off one corner and stress that patch
- 6) Deep-plough and destroy stubble after harvest

E. Fertilizer note

This is the disease where fertiliser IS the treatment. Give the full FRG-2018 dose, especially potash (MoP) and, on old rice soils, zinc sulphate at 8-10 kg/ha and gypsum for sulphur. Do not under-fertilise to save money — the yield loss from brown spot will cost far more than the fertiliser.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Propiconazole 25% EC Propiconazole	1 ml per litre of water (about 10 ml per 10 litre sprayer load)	14	2	30
2	(unverified) Mancozeb 80% WP Mancozeb	2 g per litre of water (about 20 g per 10 litre sprayer load)	10	3	21

[AP AP-374]

Regimen 1 — Timing: At first appearance of spots on the upper leaves; repeat once after 12-15 days if it keeps spreading. | Water: 4-5 litres of spray solution per decimal at full canopy | Brands: Tilt 250 EC, Propicon 250 EC | PHI note: Do not harvest for 30 days after the last spray. Propiconazole is classed as a highly hazardous pesticide in some revie...

Regimen 2 — Timing: Protective spray when weather turns humid and spots are just starting. | Water: 4-5 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 if spots cover more than 20% of leaf area, if seedlings are dying in the nursery, or if the same field gets brown spot every year — that points to a soil problem your Sub-Assistant Agriculture Officer should look at with a soil test.

III.29 Rice — Rice blast

Bangla: ধান — ধানরে ব্লাস্ট রোগ. Key: Rice__Blast. Pathogen: Magnaporthe oryzae (Pyricularia oryzae). Severity: high. Pack confidence: high.

A. Summary

Blast is the most damaging rice disease in Bangladesh and can attack the leaf, the node and the neck of the panicle. Neck blast is the dangerous stage — the panicle turns white and gives no grain. Cool nights with heavy dew, cloudy days and

too much urea make it spread fast.

B. Symptoms

- 1) Diamond or eye-shaped spots on leaves — grey/ash centre with a brown or reddish-brown border
- 2) Spots join together and whole leaves dry up in bad attacks
- 3) Black-brown ring around the node; the stem breaks easily at that point (node blast)
- 4) Blackish-brown band at the base of the panicle; the panicle stands white and upright with empty grain (neck blast)
- 5) Worst in fields given heavy urea, and after foggy nights followed by warm days

C. Immediate actions

- 1) Stop urea at once: Do not give any more urea top-dressing until the disease is under control. Extra nitrogen makes blast much worse. If you were about to top-dress, hold it back.
- 2) Drain and re-flood the field: Drain the field, let the soil crack slightly for 1-2 days, then flood again to about 5 cm. Never let a blast-affected field dry out completely — water stress makes blast worse.
- 3) Add MoP (potash): Top-dress 5-7 kg MoP per bigha (33 decimal) if you have not already applied the full potash dose. Potash helps the plant resist blast.
- 4) Spray a recommended fungicide: Spray a tricyclazole or trifloxystrobin+tebuconazole product in the late afternoon when there is no wind. Wet the leaves and the panicle base properly. Do not mix with urea in the...
- 5) Protect the panicle stage: The critical time is booting to just before flowering. If blast is anywhere in your village, spray once at booting and again at 50% heading even if your own field looks clean.

D. Prevention

- 1) Grow BRRi blast-tolerant varieties such as BRRi dhan63, BRRi dhan81, BRRi dhan89 or BRRi dhan92 in blast-prone areas
- 2) Use healthy, treated seed; do not keep seed from a blast-affected field
- 3) Apply nitrogen in 3 equal splits instead of one heavy dose
- 4) Keep 20 cm wide walking gaps every 2 metres so air moves through the field
- 5) Do not plant too densely; wider spacing dries the canopy faster
- 6) Burn or bury infected stubble after harvest; do not leave it on the bund

E. Fertilizer note

Follow the BARC FRG-2018 dose for your soil. Split urea into 3 equal top-dressings (at 15, 30 and 45 days after transplanting for Boro) rather than one heavy dose. Apply the full MoP dose — potash is the single most useful nutrient against blast. Never top-dress urea while blast is active.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Tricyclazole 75% WP Tricyclazole	0.6 g per litre of water (about 6 g per 10 litre sprayer load)	12	2	21
2	(unverified) Trifloxystrobin 25% + Tebuconazole 50% WG Trifloxystrobin + Tebuconazole	0.4 g per litre of water (about 4 g per 10 litre sprayer load)	14	2	21

Regimen 1 — Timing: Preventive spray at booting stage, and a second spray at 50% heading. For leaf blast, spray as soon as the first eye-shaped spots appear. | Water: 4-5 litres of spray solution per decimal at full canopy (about 150 litres per b... | Brands: Trooper 75 WP, Beam 75 WP, Blasticulture, Sunny 75 WP | PHI note: Do not harvest for at least 21 days after the last spray. Tricyclazole residues in rice grain are strictly limited in e...

Regimen 2 — Timing: Booting to early heading. Useful where tricyclazole has been used repeatedly, to rotate the mode of action and slow resistance. | Water: 4-5 litres of spray solution per decimal at full canopy | Brands: Nativo 75 WG | PHI note: Do not harvest for at least 21 days after the last spray.

G. Escalation

Call 16123 or go to your Upazila Agriculture Office if neck blast has appeared on more than 5% of panicles, if the disease is spreading despite two sprays, or if blast has appeared across several fields in your village. Neck blast spreads village-wide very fast and needs a coordinated response.

III.30 Rice — Rice tungro disease

Bangla: ধান — ধানরে টুংরো রোগ. Key: Rice__Tungro. Pathogen: Rice tungro bacilliform virus + Rice tungro spherical virus, transmitted by green leafhopper (Nephotettix spp.). Severity: high. Pack confidence: high.

A. Summary

Tungro is a virus carried by green leafhoppers. There is no cure once a plant is infected — no spray will bring it back. Everything you do is aimed at the leafhopper and at removing infected plants so the healthy ones survive.

B. Symptoms

- 1) Leaves turn yellow to orange-yellow, starting from the tip and moving down
- 2) Plants are visibly stunted — clearly shorter than the plants around them
- 3) Far fewer tillers than healthy hills
- 4) Young leaves may show fine rust-coloured specks or interveinal mottling
- 5) Panicles are small, poorly filled or do not emerge at all
- 6) Affected plants appear in patches, often starting at the field edge nearest a source of leafhoppers

C. Immediate actions

- 1) Pull out infected plants now: Uproot every yellow, stunted hill and carry it out of the field in a bag — do not drop it on the bund. Each infected hill is a source the leafhoppers feed on and carry from. Do th...
- 2) Check for green leafhoppers: Sweep the canopy with your hand or a net at dusk. If you find leafhoppers, the virus is still actively spreading and insecticide is justified.
- 3) Spray for the leafhopper, not the virus: Apply a recommended insecticide against green leafhopper. Understand clearly: this protects the still-healthy plants, it does not cure the yellow ones.
- 4) Do not replant into gaps: Filling gaps with fresh seedlings just gives the leafhoppers new young plants to infect. Leave the gaps for this season.
- 5) Tell your neighbours: Tungro moves between fields. Village-wide synchronous action — everyone roguing and controlling leafhoppers in the same week — works far better than one farmer acting alone.

D. Prevention

- 1) Grow tungro-tolerant BRRI varieties where tungro is a regular problem in your area
- 2) Transplant on a synchronised village-wide schedule — staggered planting keeps a continuous food supply for leafhoppers
- 3) Keep a fallow break between crops to break the leafhopper cycle
- 4) Remove weedy rice and grass weeds on bunds where leafhoppers shelter
- 5) Do not raise seedbeds next to a standing infected crop
- 6) Protect natural enemies — spiders and mirid bugs eat leafhoppers; avoid unnecessary broad-spectrum insecticide early in the season

E. Fertilizer note

Keep nitrogen at the FRG rate and split it — lush green crops attract more leafhoppers. Apply full potash. Do not try to 'feed the plant back to health' with extra urea; an infected plant will not recover and extra nitrogen only draws more leafhoppers.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Imidacloprid 20% SL Imidacloprid [unverified]	0.5 ml per litre of water (about 5 ml per 10 litre sprayer load)	14	2	21

Regimen 1 — Timing: Only when green leafhoppers are actually present. Do not spray on a calendar schedule — routine spraying kills the spiders that control leafhoppers naturally a... | Water: 4-5 litres of spray solution per decimal | Brands: Admire 20 SL, Imidor 20 SL, Tido 20 SL | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 as soon as you suspect tungro — do not wait. Tungro spreads field to field and an early village-level response saves crops. Also call if more than 5% of hills are yellow and stunted.

III.31 Rice — Sheath blight

Bangla: ধান — ধানরে খোলপোড়া রোগ। Key: Rice__Sheath_blight. Pathogen: Rhizoctonia solani. Severity: high. Pack confidence: high.

A. Summary

Sheath blight starts as greenish-grey water-soaked patches on the leaf sheath at the water line and climbs up the plant. It thrives in dense, heavily fertilised Boro crops in warm humid weather and can cut yield badly if it reaches the flag leaf.

B. Symptoms

- 1) Oval, greenish-grey water-soaked patches on the leaf sheath just above the water line
- 2) Patches enlarge, dry to straw colour with an irregular purple-brown border — like a snake-skin pattern
- 3) Lesions climb the tiller; when they reach the flag leaf sheath the panicle fills poorly
- 4) White cottony fungal threads and small brown mustard-seed-like sclerotia on the sheath in humid weather
- 5) Worst in thick stands, high nitrogen, and in the warm humid end of the Boro season

C. Immediate actions

- 1) Drain the field: Drain to alternate wetting and drying. Lowering the water line takes the fungus away from the sheath and slows it noticeably.
- 2) Stop urea: Hold back remaining nitrogen top-dress. Dense, lush, over-fertilised crops get sheath blight badly.
- 3) Open the canopy: Remove weeds and, where practical, keep walking gaps so air and sunlight reach the base of the plants.
- 4) Spray at the base, not the top: Aim the spray at the lower third of the plant where the lesions are. Spraying only the top leaves is the most common mistake and wastes the whole application.
- 5) Protect the flag leaf: If lesions have climbed above the middle of the tiller, spray immediately — once the flag leaf sheath is infected the yield loss is locked in.

D. Prevention

- 1) Do not transplant too densely — keep 20 x 15 cm or 25 x 15 cm spacing
- 2) Use alternate wetting and drying (AWD) irrigation instead of continuous flooding
- 3) Apply nitrogen in splits at the FRG rate; do not exceed it
- 4) Apply the full potash dose
- 5) Remove and destroy stubble and the floating sclerotia after land preparation — skim them off the puddled water surface
- 6) Control weeds, especially grasses that host *Rhizoctonia*

E. Fertilizer note

Do not exceed the FRG-2018 nitrogen rate and always split it. Apply the full MoP dose — potash strengthens the sheath. Avoid late nitrogen top-dressing in a crop that already looks dark green and dense.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Hexaconazole 5% EC Hexaconazole	2 ml per litre of water (about 20 ml per 10 litre sprayer load); labe...	15	2	30
2	[AP AP-460] Carbendazim 50% WP Carbendazim	1 g per litre of water (about 10 g per 10 litre sprayer load); label ...	15	2	30

[AP AP-176, AP-241]

Regimen 1 — Timing: At first appearance of sheath lesions, typically maximum tillering to booting. Repeat after 15 days if lesions keep climbing. | Water: 5 litres of spray solution per decimal, directed low into the canopy | Brands: Contaf 5 EC, Hexa 5 EC | PHI note: Do not harvest for 30 days after the last spray.

Regimen 2 — Timing: At first lesions. Rotate with hexaconazole rather than using carbendazim repeatedly — *Rhizoctonia* develops resistance to it quickly. | Water: 5 litres of spray solution per decimal, directed low | Brands: Bavistin DF, Knowin 50 WP, Autostin 50 WDG | PHI note: Do not harvest for 30 days after the last spray.

G. Escalation

Call 16123 if lesions have reached the flag leaf sheath on more than 10% of tillers, or if the disease returns every Boro season in the same plot.

III.32 Squash / Pumpkin — Powdery mildew

Bangla: কুমড়া / মশ্টি কুমড়া — কুমড়ার পাউডার মিলিডডি রোগ. Key: Squash__Powdery_mildew. Pathogen: *Podosphaera xanthii* / *Erysiphe cichoracearum*. Severity: medium. Pack confidence: medium.

A. Summary

White powdery patches on the top of the leaves, like flour dusted on them, which wipe off with a finger. It likes warm dry days with humid nights. Wettable sulphur controls it well if you start early.

B. Symptoms

- 1) White to greyish powdery patches on the upper leaf surface, stems and leaf stalks
- 2) Patches merge until the leaf looks whitewashed

- 3) Leaves yellow, dry and die from the oldest first
- 4) Fruit is not directly infected but is small, poorly coloured and sunburnt once the leaves go
- 5) Does NOT need leaf wetness — thrives in warm dry days with humid nights

C. Immediate actions

- 1) Spray wettable sulphur at the first white patches: 2 g per litre, both leaf surfaces. Early treatment is far more effective than late.
- 2) Never apply sulphur above 32°C: Cucurbits scorch badly. In hot weather use hexaconazole or difenoconazole instead.
- 3) Remove the worst old leaves: Strip whitened lower leaves and take them out of the field.
- 4) Improve air movement: Trellis vines where practical and thin dense growth.

D. Prevention

- 1) Trellis where practical; wide spacing
- 2) Remove old lower leaves regularly
- 3) Grow tolerant varieties
- 4) Avoid heavy shade
- 5) Destroy crop residue and rotate for a season

E. Fertilizer note

Follow FRG-2018 for cucurbits. Avoid excess nitrogen. Adequate potassium helps the leaves resist infection and improves fruit quality.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Sulphur 80% WP / WG	2 g per litre of water	10	4	3

Regimen AP-221: Sulphur
Regimen AP-221: At the first white patches, repeat every 10 days. | Water: 3-4 litres of spray solution per decimal | Brands: Thiovit Jet 80 WG, Microthiol Special 80 WP, Sulcox 80 WP | PHI note: Do not pick for 3 days after the last spray.

G. Escalation

Call 16123 if the whole canopy is whitening or if you cannot tell powdery from downy mildew.

III.33 Strawberry — Leaf scorch

Bangla: স্ট্রবেরি — স্ট্রবেরির পাতা বলসানো রোগ. Key: Strawberry__Leaf_scorch. Pathogen: Diplocarpon earlium. Severity: medium. Pack confidence: low.

A. Summary

Many small dark purple spots on the leaves that merge until the whole leaf looks scorched and dries up. It spreads in wet weather and is managed with sanitation, drip irrigation and a protectant fungicide.

B. Symptoms

- 1) Numerous small (1-4 mm) irregular dark purple spots on the upper leaf surface
- 2) Spots have no light centre — this distinguishes leaf scorch from leaf spot, which has a grey centre
- 3) Spots merge and the leaf takes on a scorched, purple-brown, dried appearance
- 4) Petioles, runners and fruit stalks also develop dark lesions
- 5) Heavy infection reduces fruit size and number
- 6) Worst in wet weather, dense plantings and where old leaves are left on the plant

C. Immediate actions

- 1) Remove old and scorched leaves: Pick off every heavily spotted leaf and take it out of the field. Regular deleafing is the backbone of strawberry disease management.
- 2) Switch to drip irrigation: Never irrigate strawberry overhead. Wet leaves and wet fruit cause almost every strawberry disease problem.
- 3) Mulch the beds: Straw or plastic mulch keeps leaves and fruit off the soil and stops splash-borne infection.
- 4) Spray a protectant: Mancozeb at 2 g/L or copper oxychloride at 3 g/L every 10-12 days in wet weather.
- 5) Do not take runners from infected plants: The fungus travels on planting material. Take runners only from clean mother plants.

D. Prevention

- 1) Use clean, certified runners or tissue-culture plants

- 2) Plant on raised beds with plastic mulch
- 3) Drip irrigation only
- 4) Wide spacing for air movement
- 5) Remove old leaves after each harvest flush
- 6) Rotate and destroy old plantings; do not carry plants over year after year

E. Fertilizer note

Use fertigation in small frequent doses. Keep calcium and potassium adequate for firm fruit. Avoid excess nitrogen, which produces soft dense foliage.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2 g per litre of water	10	4	7

Mancozeb

Regimen/Verification: Preventive, every 10-12 days in wet weather. Time sprays right after a picking. | Water: 2-3 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45 | PHI note: Do not pick for 7 days after the last spray. Strawberry is eaten fresh and unwashed by many consumers — the PHI matters...

G. Escalation

Call 16123 if most leaves are scorched, if fruit stalks are dying back, or if the problem returns each season on the same land.

III.34 Tomato — Bacterial spot

Bangla: টমটো — টমটোর ব্যাকটেরিয়াজনিত দাগ রোগ। Key: Tomato__Bacterial_spot. Pathogen: Xanthomonas spp. (X. euvesicatoria, X. perforans and related). Severity: medium. Pack confidence: medium.

A. Summary

This is a bacterial disease, so fungicides do not cure it. Small dark greasy spots appear on leaves and raised scabby spots on the fruit, which makes the fruit unsellable. It spreads in warm wet weather and through infected seed.

B. Symptoms

- 1) Small dark brown to black spots on leaves, often with a yellow halo, looking greasy or water-soaked when fresh
- 2) Spots may drop out leaving a shot-hole appearance
- 3) Raised, rough, scabby spots on the fruit — this is what destroys the market value
- 4) Elongated dark lesions on stems and leaf stalks
- 5) Leaves yellow and drop, exposing fruit to sunscald
- 6) Worst in warm (24-30°C) wet weather, after heavy rain or overhead irrigation

C. Immediate actions

- 1) Do not spray an ordinary fungicide expecting a cure: Bacteria do not respond to fungicides. Only copper-based products help, and they only slow the spread — they do not cure infected tissue.
- 2) Stop overhead irrigation today: The bacteria spread in splashing water. Switch to drip or furrow irrigation immediately. This is the most effective single action.
- 3) Stay out of wet plants: Do not prune, tie or pick when plants are wet. Wet hands and clothes spread bacterial spot faster than rain does.
- 4) Remove badly infected plants: Pull out and bag heavily infected plants and all spotted fruit. Bury deep or burn.
- 5) Apply copper: Copper oxychloride or copper hydroxide at 2-3 g/L every 7-10 days as a protectant on the healthy tissue.
- 6) Do not save seed from this crop: The bacteria are seed-borne. Saving seed from an infected field guarantees the same problem next year.

D. Prevention

- 1) Buy certified, treated seed; never save seed from an infected crop
- 2) Hot water seed treatment at 50°C for 25 minutes kills seed-borne Xanthomonas
- 3) Rotate away from tomato, chilli and capsicum for 2 years
- 4) Use drip or furrow irrigation only
- 5) Stake plants and mulch so fruit and leaves stay off the soil
- 6) Disinfect stakes, ties and tools between seasons

E. Fertilizer note

Avoid excess nitrogen, which produces soft tissue that infects easily. Maintain calcium and potassium at FRG rates. Do not apply foliar fertiliser sprays while the disease is active — you would be spreading bacteria in the spray water.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Copper oxychloride 50% WP Copper oxychloride	3 g per litre of water (about 30 g per 10 litre sprayer load)	7	5	7

[AP AP-354]

Regimen 1 — Timing: Every 7-10 days while warm wet weather continues. | Water: 3-4 litres of spray solution per decimal | Brands: Cupravit 50 WP, Champion 77 WP, Blue Copper | PHI note: Do not pick for 7 days after the last spray. Copper accumulates in soil — avoid using it every season on the same land.

G. Escalation

Call 16123 if fruit is developing scabby spots, if more than 20% of plants are affected, or if you are not sure whether this is bacterial spot or early blight — the management differs.

III.35 Tomato — Early blight

Bangla: টমটো — টমটোর আগাম ধ্বসা রোগ. Key: Tomato__Early_blight. Pathogen: Alternaria solani / Alternaria linariae. Severity: medium. Pack confidence: medium.

A. Summary

Brown target-like spots start on the oldest bottom leaves and climb up, stripping the plant from below. The fruit loses its shade and gets sunscald. It is slower than late blight and very manageable if you catch it on the lower leaves.

B. Symptoms

- 1) Brown spots with concentric rings inside, surrounded by a yellow halo
- 2) Always starts on the oldest leaves nearest the ground
- 3) Leaves yellow and fall; the plant becomes bare at the bottom
- 4) Dark sunken lesions with rings at the stem end of the fruit
- 5) Dark elongated lesions on the stem, sometimes girdling seedlings (collar rot)
- 6) Worse in warm humid weather and on nitrogen-starved or fruit-loaded plants

C. Immediate actions

- 1) Strip the lower leaves: Remove all spotted leaves from the bottom 30 cm of the plant and carry them out of the field. This alone slows early blight a lot and costs nothing.
- 2) Mulch the beds: Put straw or plastic mulch under the plants. Most first infections come from soil splashing onto the lowest leaves in rain or irrigation.
- 3) Feed the plant: Early blight hits hungry, stressed, heavily-fruited plants. Give a nitrogen and potassium top-dress at the FRG rate if the plant looks pale or exhausted.
- 4) Spray a protectant: Mancozeb at 2-2.5 g/L or chlorothalonil, every 7-10 days, covering the lower canopy properly. If already spreading, use difenoconazole or azoxystrobin once, then return to protect...
- 5) Keep the leaves dry: Irrigate at the base in the morning, not overhead in the evening.

D. Prevention

- 1) Rotate away from tomato, potato, brinjal and chilli for at least 2 years
- 2) Use certified disease-free seed; treat seed before sowing
- 3) Stake plants so no leaf touches the soil
- 4) Mulch every bed
- 5) Remove and destroy all crop residue at the end of the season
- 6) Do not work in the field when the plants are wet

E. Fertilizer note

Keep nitrogen and potassium adequate throughout fruiting — early blight is a stress disease. A heavy fruit load with no top-dressing is the classic trigger. Follow FRG-2018 and split the top-dressings across the picking period.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	7	6	7

Mancozeb
[unverified]

2	Copper hydroxide 77% WP Copper hydroxide (or copper oxychlo...	2 g per litre of water	10	4	7
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[AP AP-354]

Regimen 1 — Timing: Preventive, every 7-10 days from first flowering. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not pick for 7 days after the last spray.

Regimen 2 — Timing: Preventive, and acceptable in low-input or organic-leaning systems. | Water: 3-4 litres of spray solution per decimal | Brands: Champion 77 WP, Cupravit 50 WP, Blue Copper | PHI note: Do not pick for 7 days after the last spray. Copper accumulates in soil with repeated seasons.

G. Escalation

Call 16123 if defoliation reaches the middle of the plant, if fruit is developing lesions, or if seedlings are dying of collar rot in the nursery.

III.36 Tomato — Late blight

Bangla: টমটো — টমটোর নাবিধ্বসা রোগ। Key: Tomato__Late_blight. Pathogen: Phytophthora infestans. Severity: high. Pack confidence: high.

A. Summary

The same pathogen that destroys potato also attacks tomato, and it moves between the two crops. In foggy December-January weather it can kill a tomato field in about a week. Spray preventively before you see it — once the stems are black it is too late.

B. Symptoms

- 1) Large water-soaked greyish-green patches on leaves that quickly turn brown-black
- 2) White downy fungal growth on the underside of lesions in the early morning
- 3) Dark brown to black lesions on stems and leaf stalks that girdle and collapse them
- 4) Firm, greasy-looking brown patches on green fruit, often on the shoulder; the fruit stays hard, not soft and watery
- 5) Whole plants collapse within days in fog and drizzle

C. Immediate actions

- 1) Spray the same day: Do not wait. Use a systemic + contact combination (metalaxyl/benalaxyl + mancozeb, or cymoxanil + mancozeb) at the first lesion.
- 2) Remove infected plants and fruit: Cut out collapsed plants and pick off blighted fruit. Carry them out of the field in a bag and bury deep or burn. Do not throw them on the bund or feed them to cattle.
- 3) Stop wetting the leaves: Switch to furrow or drip irrigation. Never irrigate in the evening. Wet leaves overnight are exactly what the disease needs.
- 4) Open up the canopy: Remove the lower leaves touching the soil, stake the plants and prune enough to let air through. A dry canopy stops the disease better than any spray.
- 5) Check what is planted next door: If there is potato in the next field, tell that farmer. The disease will bounce between the two crops all season.

D. Prevention

- 1) Never plant tomato next to or immediately after potato
- 2) Use raised beds with good drainage and mulch
- 3) Stake and prune so the canopy dries by mid-morning
- 4) Space plants at 60 x 40 cm or wider; crowded plots blight first
- 5) Grow BARI Tomato varieties suited to your season; use polytunnels for summer tomato
- 6) Destroy all crop residue at the end of the season and rotate for 2 years

E. Fertilizer note

Avoid excess nitrogen; soft leaves blight faster. Apply the full FRG-2018 potash dose and keep calcium adequate to avoid blossom-end rot at the same time. Do not top-dress urea once blight has started.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP Mancozeb	2-2.5 g per litre of water (about 20-25 g per 10 litre sprayer load)	7	6	7
2	[AP AP-374] Benalaxyl 8% + Mancozeb 64% WP Benalaxyl + Mancozeb	2 g per litre of water (label rate approximately 0.2% of formulated p...	10	3	14

[AP AP-374]

Regimen 1 — Timing: Preventive, every 7-10 days from the start of the fog period. | Water: 3-4 litres of spray solution per decimal on a staked crop | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not pick tomatoes for 7 days after the last spray. Tomato is eaten fresh and often raw — the PHI matters more here t...

Regimen 2 — Timing: Curative — at the first visible lesion. Maximum 3 per season, always alternated with a contact fungicide. | Water: 3-4 litres of spray solution per decimal | Brands: Galben M, Ridomil Gold MZ 68 WG | PHI note: Do not pick tomatoes for 14 days after the last spray.

G. Escalation

Call 16123 if plants are collapsing, if the disease continues after two sprays, or if potato and tomato fields in your area are both affected.

III.37 Tomato — Leaf mould

Bangla: টমটো — টমটোর পাতার ছাতা রোগ. Key: Tomato__Leaf_Mold. Pathogen: Passalora fulva (Fulvia fulva / Cladosporium fulvum). Severity: medium. Pack confidence: medium.

A. Summary

Leaf mould is a humidity disease. Pale yellow patches on the top of the leaf with olive-green to brown velvety mould underneath. It is mainly a problem in polytunnels, net houses and very dense winter plots where the air does not move.

B. Symptoms

- 1) Diffuse pale yellow to greenish patches on the UPPER leaf surface, with no clear border
- 2) Olive-green, greyish or purple-brown velvety mould on the UNDERSIDE directly beneath each patch — this is the giveaway sign
- 3) Older lower leaves affected first
- 4) Leaves curl, dry and drop; heavy defoliation reduces fruit size
- 5) Almost always follows high humidity (above 85%) with poor air movement — very common under polytunnels

C. Immediate actions

- 1) Get the humidity down — this is the whole treatment: Open the polytunnel sides and ends every morning. Even in an open field, prune the lower leaves and stake the plants so air moves. Fungicide without ventilation will not fix this.
- 2) Remove the worst leaves: Strip the heavily moulded lower leaves and carry them out. Do not shake them — the spores travel on air.
- 3) Stop wetting the foliage: Switch to drip or furrow irrigation. Irrigate in the morning so any moisture dries before nightfall.
- 4) Spray if it keeps spreading: Chlorothalonil, mancozeb or a difenoconazole product at 7-10 day intervals, spraying the UNDERSIDE of the leaves where the mould actually is.

D. Prevention

- 1) Ventilate polytunnels and net houses every morning without fail
- 2) Wide spacing (60 x 45 cm or more) and staking
- 3) Prune lower leaves regularly to open the base of the canopy
- 4) Use resistant varieties where available — resistance to leaf mould is well established in tomato breeding
- 5) Avoid overhead irrigation entirely in protected cultivation
- 6) Sterilise or replace polytunnel plastic and remove all residue between crops

E. Fertilizer note

Avoid heavy nitrogen, which produces dense soft foliage and traps humidity inside the canopy. Keep potassium at the FRG rate. In polytunnels, use fertigation in small frequent doses rather than large top-dressings.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	7	5	7
2	Mancozeb Difenoconazole 25% EC Difenoconazole	0.5 ml per litre of water	12	2	7

[AP AP-384]

Regimen 1 — Timing: Preventive, every 7-10 days during humid spells. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45 | PHI note: Do not pick for 7 days after the last spray.

Regimen 2 — Timing: Curative, when mould is well established. Maximum 2 per season. | Water: 3-4 litres of spray solution per decimal | Brands: Score 250 EC | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if mould spreads to the upper canopy, if leaves are dropping heavily, or if it recurs in the same polytunnel every season — that usually means a ventilation design problem.

III.38 Tomato — Septoria leaf spot

Bangla: টমটো — টমটোর সেপ্টোরিয়া পাতার দাগ রোগ। Key: Tomato__Septoria_leaf_spot. Pathogen: Septoria lycopersici. Severity: medium. Pack confidence: medium.

A. Summary

Many small round spots with grey centres and dark edges cover the lower leaves, each with tiny black dots in the middle. It does not attack the fruit, but it strips the leaves and the exposed fruit then gets sunscald.

B. Symptoms

- 1) Many small circular spots, 2-4 mm, with grey or tan centres and dark brown margins
- 2) Tiny black pinhead dots (pycnidia) visible in the centre of older spots — look with a hand lens
- 3) Starts on the lowest leaves after the first fruit set and moves steadily upward
- 4) Leaves yellow, brown and drop; the plant is bare from below
- 5) Fruit is not directly infected, but sunscald follows the leaf loss
- 6) Spreads by rain splash, overhead irrigation and workers moving through wet plants

C. Immediate actions

- 1) Remove the spotted lower leaves: Take off every heavily spotted leaf from the bottom of the plant and carry it out of the field. Do this before spraying — you are removing the spore factory.
- 2) Mulch immediately: Straw or plastic mulch stops rain and irrigation splashing spores from the soil onto the leaves. For this disease mulch is not optional.
- 3) Stop working in wet plants: Your hands and clothes carry the spores from plant to plant. Only prune, tie or pick when the plants are dry.
- 4) Spray a protectant: Mancozeb at 2-2.5 g/L or chlorothalonil every 7-10 days, covering the lower canopy thoroughly. Keep going until the top leaves stay clean.

D. Prevention

- 1) Rotate away from tomato for 2 years; the fungus survives on residue and on solanaceous weeds
- 2) Remove and destroy all crop residue after harvest
- 3) Control solanaceous weeds (e.g. wild nightshade) around the field
- 4) Stake plants and mulch every bed
- 5) Use drip or furrow irrigation, never overhead
- 6) Use clean seed and a clean nursery bed

E. Fertilizer note

Keep the plant well fed through fruiting so it can keep producing new leaves faster than the disease removes old ones. Follow FRG-2018 and split top-dressings across the picking period.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	7	6	7
2	Mancozeb Copper oxychloride 50% WP (unverified) Copper oxychloride [AP AP-354]	3 g per litre of water	10	4	7

Regimen 1 — Timing: Preventive, every 7-10 days from first fruit set through the rainy or dewy period. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45, Manco 80 WP | PHI note: Do not pick for 7 days after the last spray.

Regimen 2 — Timing: Preventive, every 10 days in wet weather. | Water: 3-4 litres of spray solution per decimal | Brands: Cupravit 50 WP, Champion 77 WP, Blue Copper | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if defoliation reaches the middle of the plant or if you cannot tell this apart from early blight.

III.39 Tomato — Target spot

Bangla: টমটো — টমটোর টার্গেট স্পট রোগ। Key: Tomato__Target_Spot. Pathogen: Corynespora cassiicola. Severity: medium. Pack confidence: medium.

A. Summary

Small brown spots with light centres and faint rings, on leaves, stems and fruit. It looks a lot like early blight but the rings are fainter and it attacks the fruit more often. It is a warm humid weather disease.

B. Symptoms

- 1) Small brown spots that enlarge to 5-10 mm with a light tan centre and faint concentric rings
- 2) Spots on leaves, leaf stalks, stems AND fruit — the fruit involvement is the main clue
- 3) Sunken circular lesions with cracked centres on green and ripening fruit
- 4) Heavy defoliation in humid weather
- 5) Common in warm humid conditions and in dense unpruned plantings

C. Immediate actions

- 1) Prune and open the canopy: Remove the lower and inner leaves so air moves and the canopy dries. This is the fastest way to slow it.
- 2) Remove spotted fruit and leaves: Pick off and bag affected fruit and heavily spotted leaves. Fruit lesions do not heal and will rot in transport.
- 3) Spray a protectant: Mancozeb or chlorothalonil every 7-10 days. If already on the fruit, use a difenoconazole or azoxystrobin product once, then return to protectants.
- 4) Switch to drip or furrow irrigation: Stop wetting the foliage. Irrigate in the morning.

D. Prevention

- 1) Prune and stake to keep the canopy open
- 2) Rotate away from tomato and cucurbits for 2 years — *Corynespora* has a wide host range
- 3) Use certified seed and clean nursery beds
- 4) Mulch to reduce soil splash
- 5) Remove all crop residue at the end of the season
- 6) Avoid overhead irrigation

E. Fertilizer note

Keep nitrogen moderate — dense soft canopies trap humidity. Maintain potassium at the FRG rate to firm up fruit skin.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Mancozeb 80% WP	2-2.5 g per litre of water	7	5	7
2	Mancozeb, Azoxystrobin 25% SC (unverified), Azoxystrobin	1 ml per litre of water (about 10 ml per 10 litre sprayer load)	12	2	7

[AP AP-2312, AP-3153]

Regimen 1 — Timing: Preventive, every 7-10 days in humid weather. | Water: 3-4 litres of spray solution per decimal | Brands: Dithane M-45, Indofil M-45 | PHI note: Do not pick for 7 days after the last spray.

Regimen 2 — Timing: Curative, when lesions are on the fruit. Maximum 2 per season and always alternate with a different mode of action — strobilurin resistance develops quickly. | Water: 3-4 litres of spray solution per decimal | Brands: Amistar 25 SC, Sitro 25 SC | PHI note: Do not pick for 7 days after the last spray.

G. Escalation

Call 16123 if fruit lesions appear, if defoliation is heavy, or if you cannot distinguish this from early blight.

III.40 Tomato — Tomato mosaic virus (ToMV / TMV)

Bangla: টমটো — টমটোর মোজাইক ভাইরাস রোগ. Key: Tomato__Tomato_mosaic_virus. Pathogen: Tomato mosaic virus / Tobacco mosaic virus. Severity: medium. Pack confidence: medium.

A. Summary

A virus that makes light and dark green mottled patterns on the leaves. There is no cure. It spreads mainly by hands, tools and tobacco — not by insects — so the control is hygiene, not spraying.

B. Symptoms

- 1) Light green and dark green mottled (mosaic) pattern on the leaves
- 2) Leaves may be narrow, distorted or fern-like
- 3) Plants mildly to moderately stunted; growth slows
- 4) Uneven ripening and internal browning of fruit; yellow blotches on fruit

- 5) No insects associated — this distinguishes it from TYLCV, which comes with whitefly
- 6) Often appears in a row or line across the field, following the path a worker's hands took

C. Immediate actions

- 1) Wash your hands before touching any more plants: Wash with soap, then with milk or a 3% trisodium phosphate solution. The virus survives on skin and clothes for a long time and this is how it moves down a row.
- 2) No tobacco in the field — ever: Do not smoke bidi or cigarettes, and do not chew tobacco or jarda, while working with tomato. Tobacco products carry the virus and this is a leading cause of ToMV in Bangladesh.
- 3) Pull out infected plants: Bag and remove mottled plants. Wash your hands afterwards before touching healthy plants.
- 4) Do not spray for it: There is no chemical treatment for a plant virus. Do not buy any product sold as a 'virus cure' — no such registered product exists.
- 5) Disinfect tools and stakes: Dip pruning knives and shears in a 1% bleach or 3% trisodium phosphate solution between plants.

D. Prevention

- 1) Use certified virus-free seed; hot water or trisodium phosphate seed treatment removes seed-borne virus
- 2) Grow resistant varieties carrying the Tm-2a gene where available
- 3) Strict hand hygiene for everyone working in the field
- 4) Absolute ban on tobacco use in and around the tomato field
- 5) Do not transplant seedlings from a nursery where mottling has been seen
- 6) Remove all crop residue; the virus survives a long time in dry plant debris and on stakes

E. Fertilizer note

Balanced feeding at FRG rates helps the plant tolerate infection and keep producing, but it does not cure it. Avoid nitrogen excess.

F. Chemical regimens

No chemical regimen listed.

G. Escalation

Call 16123 if mottling appears in the nursery, if more than 10% of plants show mosaic, or if you cannot tell this apart from a nutrient deficiency.

III.41 Tomato — Tomato yellow leaf curl virus (TYLCV)

Bangla: টমটো — টমটোর পাতা কঁকড়ানো হলুদ ভাইরাস রোগ. Key: Tomato__Tomato_Yellow_Leaf_Curl_Virus. Pathogen: Tomato yellow leaf curl virus, transmitted by whitefly (*Bemisia tabaci*). Severity: high. Pack confidence: medium.

A. Summary

A virus carried by whitefly. Leaves curl upward, turn yellow at the edges and the plant stops growing. There is no cure — an infected plant will never produce properly. Everything you do is aimed at the whitefly and at removing infected plants early.

B. Symptoms

- 1) Young leaves curl UPWARD and inward, cupping like a bowl
- 2) Leaf margins turn yellow; leaves become small, thick and leathery
- 3) Plant is severely stunted with short bushy internodes
- 4) Flowers drop; very little or no fruit sets after infection
- 5) If infected early the plant may never produce a marketable fruit
- 6) Whiteflies visible on the underside of leaves — shake a plant and a white cloud rises

C. Immediate actions

- 1) Pull out infected plants today: Bag each curled, stunted plant on the spot (so whiteflies do not fly off it) and remove it from the field. Every infected plant is a reservoir the whiteflies feed on and carry fro...
- 2) Attack the whitefly, not the virus: No spray cures the virus. Control the whitefly to protect the plants that are still healthy. Be clear about this so you do not waste money on 'virus medicine'.
- 3) Put up yellow sticky traps: Install 8-10 yellow sticky traps per decimal at canopy height. They catch whiteflies and, just as importantly, tell you how bad the population is.
- 4) Spray for whitefly if numbers are high: Use imidacloprid, thiamethoxam or spiromesifen against whitefly, aimed at the UNDERSIDE of leaves where they live. Rotate the chemical group each time — whitefly develops resistan...

5) Protect the nursery: Raise seedlings under a 40-50 mesh insect-proof net. Most TYLCV infection happens in the nursery or in the first 3 weeks after transplanting, before you ever see a symptom.

D. Prevention

- 1) Raise seedlings under 40-50 mesh insect-proof net — the highest-return single practice for this disease
- 2) Grow TYLCV-tolerant varieties where available
- 3) Plant a barrier crop of maize or sorghum around the plot to slow whitefly entry
- 4) Use silver/reflective plastic mulch, which repels whitefly
- 5) Rogue infected plants weekly, starting from the first week after transplanting
- 6) Do not plant tomato next to an old tomato, brinjal, chilli or cotton field

E. Fertilizer note

Balanced feeding helps healthy plants outgrow light infection, but no amount of fertiliser will recover a badly infected plant. Do not add extra urea hoping the plant will 'come back' — soft new growth just attracts more whitefly.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Imidacloprid 20% SL Imidacloprid	0.5 ml per litre of water (about 5 ml per 10 litre sprayer load)	12	2	5
2	[unverified] Thiamethoxam 25% WG Thiamethoxam	0.2 g per litre of water (about 2 g per 10 litre sprayer load)	12	2	5

Regimen 1 — Timing: Only when whitefly numbers are high on the sticky traps. Do not spray on a fixed calendar — that only breeds resistant whitefly. | Water: 3-4 litres of spray solution per decimal | Brands: Admire 20 SL, Imidor 20 SL, Tido 20 SL | PHI note: Do not pick for 5 days after the last spray.

Regimen 2 — Timing: Rotation partner for imidacloprid. Note both are neonicotinoids — for a true rotation, alternate with a different group such as spiromesifen or a soap/oil treat... | Water: 3-4 litres of spray solution per decimal | Brands: Actara 25 WG | PHI note: Do not pick for 5 days after the last spray.

G. Escalation

Call 16123 as soon as you see the first curled stunted plants. Early village-level action on whitefly is the only thing that limits TYLCV. Also call if more than 10% of plants are affected within 4 weeks of transplanting.

III.42 Tomato — Two-spotted spider mite

Bangla: টমটো — টমটোর লাল মাকড় (দুই ফোঁটা মাকড়). Key: Tomato__Spider_mites_Two-spotted_spider_mite. Pathogen: Tetranychus urticae (a mite, not a disease). Severity: medium. Pack confidence: medium.

A. Summary

These are tiny mites, not a disease. They live on the underside of leaves and suck sap, leaving fine pale speckling and later a silvery bronze look and fine webbing. They explode in hot dry weather and after broad-spectrum insecticide sprays that kill their natural enemies.

B. Symptoms

- 1) Fine pale yellow or white speckling (stippling) on the upper leaf surface
- 2) Leaves later turn bronze, silvery or reddish and become dry and brittle
- 3) Very fine webbing on the underside of leaves and between leaflets in heavy attacks
- 4) Tiny moving specks on the leaf underside — hold a white paper under the leaf and tap; the moving dots are mites
- 5) Starts in the hot dry patches of the field, often near a dusty path or field edge
- 6) Explodes after a broad-spectrum insecticide spray (predatory mites are killed first)

C. Immediate actions

- 1) Stop broad-spectrum insecticides: If you have been spraying cypermethrin or similar, stop. Pyrethroids kill the predatory mites that keep spider mites down and are a common cause of mite outbreaks.
- 2) Raise the humidity around the plants: Irrigate properly and, in small plots, wash the leaf undersides with a strong jet of plain water in the morning. Spider mites hate moisture and this alone knocks the population down...
- 3) Remove the worst leaves: Strip and bag the heavily webbed and bronzed leaves. Do not shake them across the field.
- 4) Use a specific miticide, not an insecticide: Ordinary insecticides do not kill mites. Use abamectin, spiromesifen or sulphur. Aim entirely at the leaf UNDERSIDE.
- 5) Rotate the miticide: Mites develop resistance within a season. Never use the same miticide twice in a row.

D. Prevention

- 1) Keep the field free of dust — dusty conditions favour mites
- 2) Irrigate adequately; drought-stressed plants get mites badly
- 3) Avoid unnecessary pyrethroid sprays that destroy predatory mites
- 4) Remove and destroy crop residue and weed hosts at the field edge
- 5) Do not plant tomato next to a mite-infested crop such as brinjal, bean or cucurbits
- 6) Scout leaf undersides weekly with a hand lens during hot dry spells

E. Fertilizer note

Avoid excess nitrogen — soft, high-nitrogen foliage lets mite populations grow much faster. Keep the plant adequately watered and fed at FRG rates; drought stress is the main trigger.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Abamectin 1.8% EC Abamectin	0.5 ml per litre of water (about 5 ml per 10 litre sprayer load)	7	2	7
2	Sulphur 80% WP (wetttable sulphur) Sulphur [AP AP-252]	2 g per litre of water (about 20 g per 10 litre sprayer load)	10	3	3

Regimen 1 — Timing: When mites are confirmed on the leaf underside. Repeat once after 7 days to catch newly hatched mites, then switch chemical group. | Water: 3-4 litres of spray solution per decimal | Brands: Vertimec 1.8 EC, Abamec 1.8 EC | PHI note: Do not pick for 7 days after the last spray.

Regimen 2 — Timing: Early infestation, or as a rotation partner. Also suppresses powdery mildew at the same time. | Water: 3-4 litres of spray solution per decimal | Brands: Thiovit Jet 80 WG, Microthiol Special 80 WP, Sulcox 80 WP | PHI note: Do not pick for 3 days after the last spray.

G. Escalation

Call 16123 if leaves are bronzing across the field, if webbing is visible, or if mites return within 2 weeks of a miticide spray — that means resistance and you need a different group.

III.43 Wheat — Wheat blast

Bangla: গম — গমের ব্লাস্ট রোগ. Key: Wheat__Blast. Pathogen: Magnaporthe oryzae Triticum pathotype (MoT). Severity: high. Pack confidence: medium.

A. Summary

Wheat blast first appeared in Bangladesh in 2016 and is a national quarantine-level concern. The spike turns completely white and bleached above the point of infection while the leaves stay green, and the grains are shrivelled or absent. Warm humid weather at heading is what triggers it.

B. Symptoms

- 1) The spike bleaches to a straw-white colour ABOVE a dark lesion on the rachis, while the part below stays green
- 2) Leaves often remain green — this is the classic contrast that identifies wheat blast
- 3) A dark brown to black band at the point of infection on the spike stem
- 4) Grains are shrivelled, small, discoloured or completely absent
- 5) Whole fields can turn white within days at heading in warm humid weather
- 6) Diamond-shaped grey lesions with dark borders may appear on leaves

C. Immediate actions

- 1) Report it — this is a notifiable-level concern: Call the Krishi Call Center on 16123 and inform your Upazila Agriculture Office immediately. Wheat blast is under national surveillance in Bangladesh and early reporting helps con...
- 2) Do not save seed from this field: The pathogen is seed-transmitted. Seed from a blast-affected field will carry it to new areas. This is the single most important thing you can do.
- 3) Spray only if the crop is not yet fully bleached: A tricyclazole or trifloxystrobin+tebuconazole spray at heading can protect spikes not yet infected. Once a spike is white it cannot be saved.
- 4) Do not move the crop residue to a new area: Burn or deep-bury residue in place. Do not sell or transport straw from an affected field.
- 5) Plan an early-sown, short-duration variety next year: Sowing early so heading finishes before the warm humid window is the most reliable protection available.

D. Prevention

- 1) Sow early (mid-November) so heading finishes before mid-February warmth and humidity
- 2) Grow BARI-released blast-tolerant varieties such as BARI Gom 33 where available
- 3) Use certified seed only — never seed saved from a blast area
- 4) Rotate away from wheat in affected fields
- 5) Destroy wheat residue after harvest
- 6) Avoid wheat next to a previously affected field

E. Fertilizer note

Follow FRG-2018 for wheat. Avoid excess nitrogen, which delays maturity and pushes heading into the risky warm humid window. Apply the full potash dose.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Tricyclazole 75% WP	0.6 g per litre of water	10	2	21
2	Tricyclazole + Trifloxystrobin 25% + Tebuconazole 50% WG Trifloxystrobin + Tebuconazole [unverified]	0.4 g per litre of water	10	2	21

Regimen 1 — Timing: Preventive at heading (when 50% of spikes have emerged) and again 7-10 days later if warm humid weather continues. | Water: 4-5 litres of spray solution per decimal | Brands: Trooper 75 WP, Beam 75 WP, Sunny 75 WP | PHI note: Do not harvest for 21 days after the last spray.

Regimen 2 — Timing: At heading, as the main preventive spray. Widely recommended internationally for wheat blast suppression. | Water: 4-5 litres of spray solution per decimal | Brands: Nativo 75 WG | PHI note: Do not harvest for 21 days after the last spray.

G. Escalation

Call 16123 immediately on first suspicion. Do not wait to confirm. Early district-level reporting is the main national defence against wheat blast.

III.44 Wheat — Wheat leaf rust (brown rust)

Bangla: গম — গমের পাতার মরচি রোগ. Key: Wheat__Leaf_rust. Pathogen: Puccinia triticina. Severity: medium. Pack confidence: medium.

A. Summary

Small round orange-brown pustules scattered on the upper leaf surface that leave orange powder on your fingers. It is common in Bangladesh's Rabi wheat and is easy to manage if you spray before it reaches the flag leaf.

B. Symptoms

- 1) Small round to oval orange-brown pustules, mostly on the UPPER leaf surface
- 2) Orange powder rubs off on your fingers or clothes
- 3) Pustules are scattered randomly, not in lines
- 4) Leaves yellow and dry from the bottom up in heavy attacks
- 5) Grains are shrivelled and test weight drops
- 6) Favoured by 15-22°C with dew; common in the middle of the Bangladesh Rabi season

C. Immediate actions

- 1) Check whether the flag leaf is affected: The flag leaf and the leaf below it fill the grain. If rust is only on the lower leaves and the crop is close to maturity, spraying may not pay.
- 2) Spray if rust is on or approaching the flag leaf: Propiconazole or tebuconazole at label rate, one spray, targeting the upper leaves.
- 3) Do not add late nitrogen: It delays maturity and extends the period the rust can damage the crop.
- 4) Note the variety for next season: If your variety rusts badly every year, switch to a BARI-released rust-resistant variety — rust resistance breaks down over time and varieties need refreshing.

D. Prevention

- 1) Grow BARI-released rust-resistant wheat varieties and change them every few years
- 2) Sow on time (mid-November) so grain filling finishes before rust peaks
- 3) Use certified seed
- 4) Do not over-apply nitrogen

- 5) Remove volunteer wheat plants that carry rust between seasons
- 6) Apply the full potash dose

E. Fertilizer note

Follow FRG-2018 for wheat with the full potash dose. Avoid heavy or late nitrogen.

F. Chemical regimens

#	Product / AI	Dose	Interval	Max	PHI
1	Propiconazole 25% EC	1 ml per litre of water	15	2	30

Propiconazole
Regimen verified: When rust reaches the leaf below the flag leaf, typically at booting to early heading. One spray is usually enough. | Water: 4-5 litres of spray solution per decimal | Brands: Tilt 250 EC, Propicon 250 EC | PHI note: Do not harvest for 30 days after the last spray.

G. Escalation

Call 16123 if rust covers the flag leaf, if leaves are drying across the field, or if the spike is bleaching white — that last one may be wheat blast, which is far more serious.

IV. References

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V. Authorization

This report contains 44 disease protocols submitted for technical verification. By signing below, the reviewing officer confirms that reviewed regimens are suitable for advisory use subject to any written corrections attached to this document, and that farmers must follow registered product labels and local official guidance.

Official verification

Reviewer name: _____

Designation / office: _____

Organization: _____

Contact phone: _____

Date: _____ Place: _____

Decision: ☐ Approved ☐ Approved with corrections ☐ Not approved

Signature: _____ Official seal:

